
Airline customer satisfaction: a bayesian prediction model

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Abstract

In this project we define and fit a Bayesian prediction model for the satisfaction level of customers of an undisclosed airline, based on a set of flight characteristics. Model parameters are derived using MCMC methods via the JAGS tool. We compare a logistic regression under weakly informative priors against a horseshoe shrinkage prior over 22 predictors, assess both with WAIC and PSIS-LOO, and validate the selected model on a held-out set of 128274 customers.

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1. Problem description and data exploration

1.1 Problem description

The goal of this project is to predict the level of customer satisfaction of an unspecified airline, given a set of covariates X that describe several characteristics of the flight, such as service quality items, type of customer, departure delay and so on, and customer's opinion on the flight, like the rating given to some service quality items. Given the high degree of uncertainty inherent in personal opinions, we chose to adopt a Bayesian logistic regression model[1]. The Bayesian coefficients are estimated via Markov Chain Monte Carlo (MCMC) methods, sampled in JAGS.

1.2 Dataset description

The data set [2] contains 129,880 customer records, described by 22 columns. Arrival Delay is missing for 393 records (0.3% of the data); given the negligible share, we drop these rows rather than impute them, leaving 129,487 clean records. Of these, a stratified subsample of 1213 rows is used for model fitting (Section 1.3), and the remaining 128,274 records are held out for out-of-sample validation (Section 6).

The columns are organized as follows:

- **Target variable:** represents the satisfaction level of customer i , denoted y_i , which can be either *dissatisfied* or *satisfied* and is encoded respectively as $\{0, 1\}$.
- **Categorical variables (3):** indicate the category of the customer and of the flight to which the record refers. For these variables, one-hot encoding is used. Variable list: Customer type, Type of Travel, Class.
- **Discrete variables (15):** mostly composed of quality-of-service variables. Variable list (service-quality items excluded): Age.
 - **Service-quality variables (14):** indicate the various levels of service available throughout the whole journey, from booking to arrival. These variables contain several zeros, which may indicate that the corresponding service was not available during the flight. Variable list: Seat comfort, Departure/Arrival time convenience, Food and drink, Gate location, Inflight wifi service, Inflight entertainment, Online support, Ease of Online booking, On-board service, Leg room service, Baggage handling, Check-in service, Cleanliness, Online boarding.
- **Flight properties (3):** report the spatial and temporal properties of the flight. They are treated as continuous variables. Variable list: Flight distance, Departure delay, Arrival delay.
- **Modified variables:**
 - owing to the extremely high correlation between Departure delay and Arrival delay, we decided to replace the latter with the difference between the two delays, so as to retain the information while reducing the correlation.
 - given the high skewness of variables Departure Delay and Mid-flight delay, we decided to correct it through a logarithm transformation. Flight Distance was tested but kept on its original scale as transforming it worsened rather than improving the prediction.

Further details on missing-data handling and the delay transformations are given in Appendix A.

1.3 Dataset exploration

We begin our exploration by examining the balance of the target variable for the prediction model. The two classes are close to even, 54.7% satisfied against 45.3% dissatisfied (Tab. 1), so the use of models more complex than a Bernoulli probability distribution is not justified and will not be pursued.

To obtain a first indication of the relationship between the response variable and the remaining covariates, we examine their correlation, as shown in Fig. 1a. Focusing on the service-quality variables (Fig. 1b), we can distinguish three subgroups of features. The correlation is computed using R's `corr()` function, which is based on Pearson's correlation coefficient

$$r_{X,Y} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Tab. 1 reports the class balance of the remaining categorical variables. Given the class imbalance of some covariates, we decided to stratify the train–test split of the dataset.

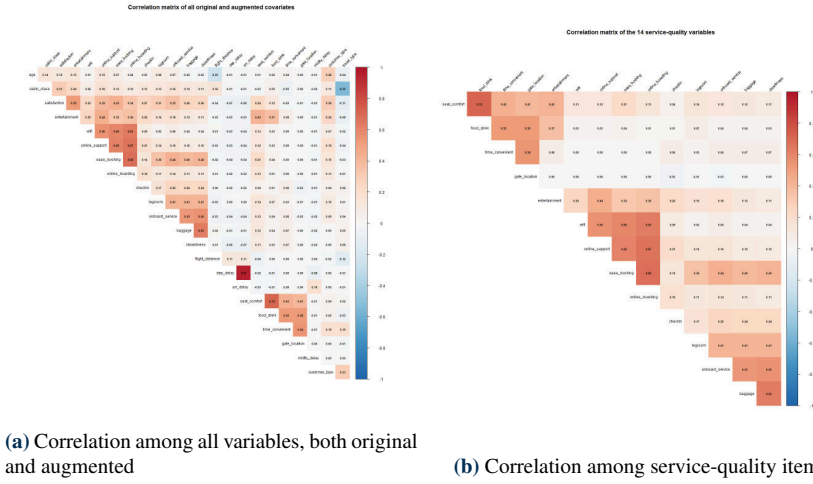


Figure 1. Correlation structure and class balance of the covariates

Table 1. Class balance of the categorical covariates

Variable	Level	Prop.
Satisfaction	Dissatisfied	0.453
	Satisfied	0.547
Customer type	Disloyal customer	0.183
	Loyal customer	0.817
Travel type	Business travel	0.691
	Personal travel	0.309
Class	Eco	0.449
	Eco Plus	0.072
	Business	0.479

2. Model specification

Let y_i denote the satisfaction level of customer i . We assume that

$$y_i \mid p_i \sim \text{Bernoulli}(p_i)$$

where p_i is the probability that customer i is satisfied. We model the probability p_i as a function of the covariates X as

$$\text{logit}(p_i) = \ln \left(\frac{p_i}{1 - p_i} \right) = \beta_0 + x_i \beta + \beta_{\text{mid}} \cdot \text{midflight_delay}_i.$$

where β_0 is the intercept in the logit space and β is the vector of regression coefficients of the remaining original covariates X (after skewness adjustment, eventual hot-encoding for categorical variables and data augmentation), x_i is the covariates set of customer i . The mid-flight delay is deliberately kept outside the main design matrix and given its own coefficient β_{mid} , paired with an explicit per-row sampling statement $\text{midflight_delay}_i \sim \mathcal{N}(0, 1)$ in the JAGS model.

Later in the project we will carry out the analysis by using several models, and for each one of them we'll describe the structure of the prior we decide to use for the regression coefficients.

3. Logit model

In this chapter we present the results obtained by fitting a simple Bayesian logistic regression on a reduced set of covariates, in order to establish a first, fully interpretable baseline solution and to settle a few modelling choices (prior specification, class encoding, MCMC setup) reused by the shrinkage model of Section 4.

3.1 Covariates and prior specification

Following the suggestions of the brief, we fit the model on a compact subset of covariates, using the transformations motivated in the data-exploration phase (Section 1.3): age, seat_comfort, flight_distance, class, the log-transformed departure delay $\log(1 + \text{dep_delay})$ and the signed-log mid-flight delay, together with the two binary descriptors travel_type and customer_type. All continuous covariates are standardized (zero mean, unit variance) so that the common prior variance is equally informative for every covariate. For the binary covariates the reference level is chosen as the majority group (*Business travel* and *Loyal customer*). The model 2 is still the same, just reduced at these variables.

Prior specification. Since we have no strong prior belief on the magnitude or the sign of the coefficients, and because standardisation already puts every covariate on a comparable scale, we adopt independent, weakly informative Gaussian priors on all the parameters:

$$\beta_0 \sim \mathcal{N}(0, 100), \quad \beta_j \sim \mathcal{N}(0, 100) \quad (j = 1, \dots, P), \quad \beta_{\text{mid}} \sim \mathcal{N}(0, 100). \quad (1)$$

Here the second argument denotes the prior variance $\sigma^2 = 100$ (equivalently a precision $1/\sigma^2 = 0.01$ in the JAGS parametrisation). On the logit scale a standard deviation of 10 is very diffuse: it places essentially no informative constraint on plausible coefficient values while still being proper, so the posterior is driven mostly by the likelihood.

3.2 Encoding of the flight class

Before committing to the model we address a small preliminary question: is it preferable to encode `class` as a pair of dummy variables (*Eco Plus* and *Business*, with *Eco* as reference) or as a single centred ordinal score (1/2/3)? To decide, we fit the model (1) under both encodings and compare them with the Widely Applicable Information Criterion (WAIC) [1], computed from the pointwise log-likelihood of the MCMC draws.

The dummy encoding attains the lower WAIC (1373.4 against 1379.1 for the ordinal one), i.e. a slightly better expected predictive fit. The difference in expected log predictive density is $\Delta\text{elpd} = -2.8$ with a standard error of 2.7, so the preference, although real, is smaller than two standard errors and therefore modest. Because `class` is genuinely *nominal* rather than an equally-spaced ordinal scale, and because the dummy specification is both more flexible and lets us report a separate, directly interpretable odds ratio for each class, we adopt the dummy encoding for the rest of the analysis. To verify that this in-sample preference is not an artefact of the modest training size ($n = 1213$), we re-score both encodings on the held-out test set ($n \approx 128,000$ customers), using the posterior-mean predictive probability $\hat{p}_i = S^{-1} \sum_{s=1}^S \text{logit}^{-1}(\hat{\beta}_0^{(s)} + x_i \hat{\beta}^{(s)})$ obtained from the already-fitted MCMC draws (without additional sampling). The dummy encoding again outperforms the ordinal one on all three out-of-sample scoring rules considered: log-loss (0.557 against 0.562), Brier score (0.187 against 0.190), and AUC (0.786 against 0.781).

3.3 MCMC setup and convergence

Posterior inference is carried out via MCMC in JAGS. Table 2 summarises the choices on the chain parameters.

Setting	Value
Chains	3
Adaptation	1500
Burn-in	1000
Iterations	12000
Thinning	7
Retained draws	5142

Table 2. MCMC settings used for the logit model

Convergence was satisfactory for all parameters. The Gelman–Rubin [1] potential scale reduction factors are essentially 1 (point estimates 1, upper confidence limits capped at 1.01, multivariate PSRF = 1), the trace plots show well-mixed, stationary chains, and the autocorrelation decays quickly to zero within a few, small lags. The effective sample sizes are at or near the number of retained draws for most parameters, with the lowest at 2057: the Monte Carlo standard error is therefore at most 2.2% of the posterior standard deviation, so the summaries we report are stable to the precision at which we quote them. The complete convergence assessment (trace plots, autocorrelation functions and the full diagnostic tables) is deferred to Appendix B.

3.4 Posterior results

The posterior densities of the coefficients and the odds-ratio forest plot are collected in Figure 2; the corresponding numerical summaries are reported in Table 3. For each coefficient we report the posterior mean, the 95% credible interval on the log-odds scale (β) and the same quantities transformed to the odds-ratio scale ($\text{OR} = e^\beta$), which is the natural scale for interpretation.

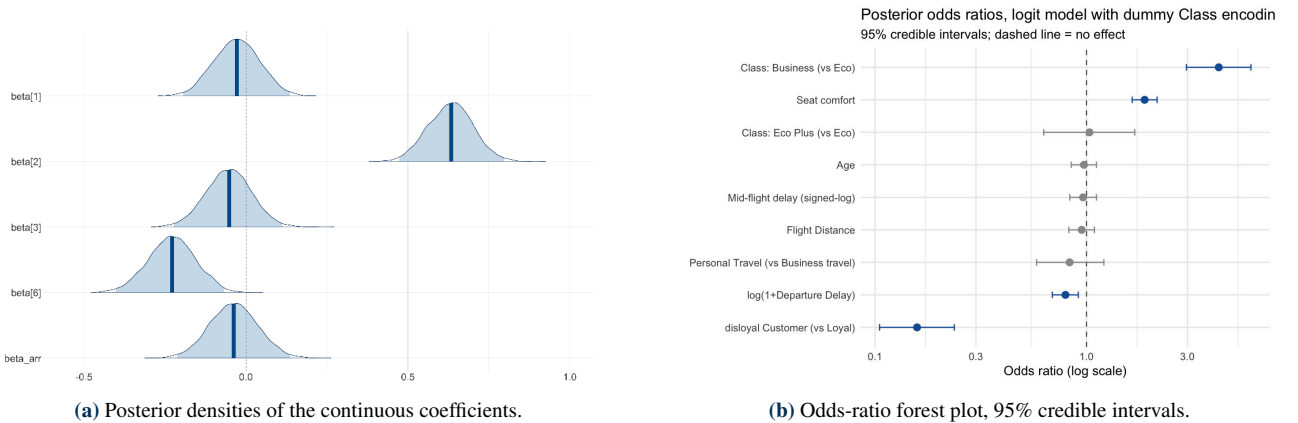


Figure 2. Baseline logit model: graphical results.

Four effects clearly separate from the “no-effect” line ($\text{OR} = 1$). The largest in magnitude is customer loyalty, and it runs negative: a *disloyal customer* has markedly lower odds of being satisfied. Cabin class comes next: travelling in *Business* rather than *Eco* multiplies the odds by roughly four, even after adjusting for all the other covariates. *Seat comfort* follows: more comfortable seating is strongly associated with satisfaction. A longer departure delay reduces the odds too.

The remaining covariates have credible intervals that comfortably include 1 and cannot be called confident drivers on their own: *Eco Plus* is statistically indistinguishable from *Eco*, while *age*, *flight distance* and the *mid-flight delay* all show weak, uncertain effects.

Table 3. Posterior summaries of the logit coefficients and their odds ratios, sorted by decreasing OR. CIs are 95% credible intervals.

Term	$\bar{\beta}$	$\beta_{2.5\%}$	$\beta_{97.5\%}$	OR	OR _{2.5%}	OR _{97.5%}
Class: Business (vs Eco)	1.440	1.088	1.791	4.221	2.967	5.993
Seat comfort	0.632	0.497	0.769	1.882	1.644	2.157
Class: Eco Plus (vs Eco)	0.032	-0.467	0.525	1.033	0.627	1.691
Age	-0.028	-0.166	0.108	0.972	0.847	1.114
Mid-flight delay (signed-log)	-0.038	-0.182	0.108	0.963	0.833	1.114
Flight distance	-0.053	-0.192	0.086	0.949	0.825	1.090
Intercept	-0.087	-0.405	0.233	0.917	0.667	1.262
Personal Travel (vs Business travel)	-0.184	-0.546	0.189	0.832	0.580	1.208
log(1 + Departure Delay)	-0.229	-0.373	-0.089	0.795	0.689	0.914
disloyal Customer (vs Loyal)	-1.845	-2.257	-1.439	0.158	0.105	0.237

4. Horseshoe shrinkage model

4.1 Motivation

The logit model of Section 3 was fit on the small set of covariates suggested by the brief, taking their usefulness for granted. That is a reasonable baseline, but it leaves open the central question of the analysis: among *all* the available predictors, in particular the fourteen service-quality ratings that were set aside earlier, which ones actually carry signal for customer satisfaction, and which are redundant? Rather than answering this by hand, adding and removing covariates and guessing at their relevance, we prefer fitting a Bayesian logistic regression on the *full* set of covariates under a **horseshoe** shrinkage prior [1], which is designed to pull the coefficients of noise predictors strongly towards zero while leaving genuine signals almost untouched. This lets the data, rather than our prior guesses, single out the meaningful covariates.

4.2 Prior specification

We include all 21 standardised predictors: the 14 service-quality items plus age, flight_distance, the two class dummies (*Eco Plus*, *Business*), log(1 + dep_delay), *Personal Travel* and *disloyal Customer*.

The horseshoe prior replaces the common vague Gaussian of Section 3 with a scale-mixture of normals: each coefficient gets its own *local* scale λ_j , and all coefficients share a single *global* scale τ ,

$$\beta_j \mid \lambda_j, \tau \sim \mathcal{N}(0, \lambda_j^2 \tau^2), \quad \lambda_j \sim \mathcal{C}^+(0, 1), \quad \tau \sim \mathcal{C}^+(0, 1), \quad (2)$$

where $\mathcal{C}^+(0, 1)$ denotes the standard half-Cauchy distribution. The global scale τ pulls *all* coefficients towards zero, enforcing overall sparsity, while the heavy Cauchy tails of the local scales λ_j let individual coefficients escape that shrinkage when the data demand it – this is precisely the “a few large, most near zero” behaviour that makes the prior suited to feature selection. The arrival-delay coefficient β_{mid} is given its own local scale λ_{mid} under the *same* global τ , so that it competes for signal on an equal footing with the other 21 predictors. Following the JAGS notes, each half-Cauchy is implemented as a Student- t with one degree of freedom truncated to the positive reals, $\text{dt}(0, 1, 1) \text{ T}(0, \cdot)$. The intercept keeps a diffuse Gaussian prior, $\beta_0 \sim \mathcal{N}(0, 100)$, and is deliberately left out of the shrinkage so that the baseline log-odds is not penalised.

A convenient by-product of the mixture form is the *shrinkage weight*

$$\kappa_j = \frac{1}{1 + \lambda_j^2}, \quad \kappa_j \in (0, 1), \quad (3)$$

which measures how much coefficient j is pulled towards zero: κ_j near 0 marks a preserved *signal* (essentially unshrunk), whereas κ_j near 1 marks a coefficient shrunk towards *noise*. We use the posterior distribution of the κ_j as our covariate-relevance summary.

4.3 MCMC setup and convergence

Because of the heavy-tailed half-Cauchy hyperpriors, the horseshoe posterior mixes more slowly than the vague-prior model of Section 3, and it is computationally heavier (22 coefficients, 22 local scales and one global scale, against the 9 parameters of Section 3). We therefore budget a longer sampling run and thinning from the outset; the settings are reported in Table 4.

4.4 Posterior results

Figure 3c shows the marginal posterior densities of the coefficients: most concentrate tightly around zero, with only a handful clearly displaced, which is the visual signature of the shrinkage at work. The relevance ranking is made explicit by the shrinkage weights κ_j of Eq. (3), plotted in Appendix D: predictors on the “signal” side (κ_j small) are the ones the model retains, while those pushed to $\kappa_j \rightarrow 1$ are treated as redundant. Figure 23, in Appendix D, reports the same information on the coefficient scale, as a forest plot of the posterior means and 95% credible intervals of the β_j , colouring in the covariates identified as signal ($\kappa_j < 0.5$).

Setting	Value
Chains	3
Adaptation	5000
Burn-in	1000
Iterations	30000
Thinning	3
Retained draws	30000

Table 4. MCMC settings used for the horseshoe model

Convergence was assessed separately for the two blocks of parameters, the regression coefficients $\{\beta_j, \beta_0, \beta_{\text{mid}}\}$ and the scale parameters $\{\lambda_j, \lambda_{\text{mid}}, \tau\}$, since the latter are the harder to sample. For each block we inspected the Gelman–Rubin potential scale reduction factors, the effective sample sizes, and the trace and autocorrelation plots. The coefficients mix well: every one has a scale-reduction factor of 1.00 and an effective sample size in the thousands. The local scales λ_j mix more slowly and show higher autocorrelation, as expected for a horseshoe, and three of them sit at or above the conventional 1.1 threshold; the global scale τ , by contrast, is among the better-behaved parameters in that block. Figures 3a and 3b show a representative trace plot for each block. The complete diagnostic material – all trace and autocorrelation plots for both blocks, together with the full Gelman–Rubin and effective-sample-size tables – is collected in Appendix C, where the three imperfectly converged scales are identified and their consequences discussed.

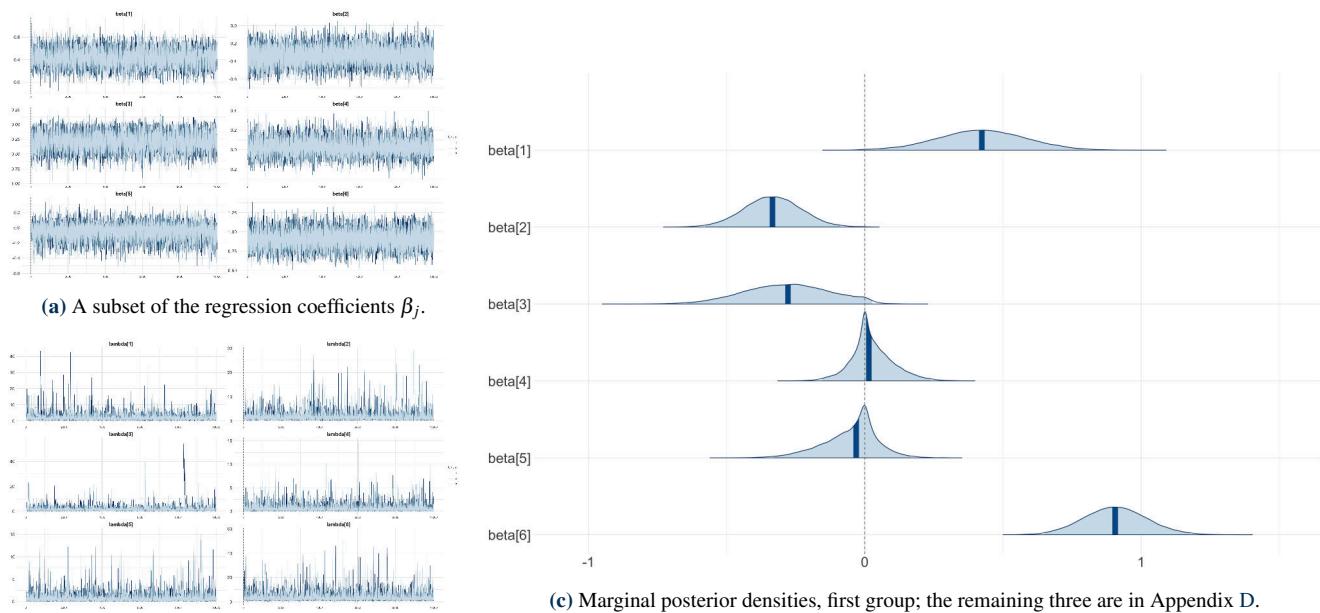


Figure 3. Representative trace plots under the horseshoe prior (three chains, left) and the resulting marginal posteriors (right).

4.5 Conclusions

The horseshoe cleanly separates the informative predictors from the redundant ones, and several of its findings sharpen – and in one case reverse – the picture from Sections 1.3 and 3.

The booking / connectivity cluster visibly competes. Among the four mutually correlated “booking / connectivity” items identified in the exploratory analysis (Section 1.3), the prior does not treat them symmetrically: *Ease of Online booking* and *Online support* are preserved noticeably more than *Online boarding* and *Inflight wifi service*, both of which sit among the most heavily shrunk items in the model ($\kappa_j = 0.74$ and 0.73). This is exactly the mechanism the horseshoe was chosen for: when several correlated items carry overlapping information, the prior lets two of them represent the signal and treats the other two as largely redundant, instead of forcing us to pick one by hand.

Business class shrinks once service quality is in the model, but remains one of the strongest effects. In Section 3, Business class alone carried a large, confident odds ratio (about 4). Here, with all 14 service ratings included, its coefficient drops to approximately 0.9 on the log-odds scale ($\kappa = 0.183$), with a 95% credible interval comfortably above zero (roughly $[0.4, 1.3]$). Together with *disloyal Customer* and *Inflight entertainment*, it remains among the three least-shrunk, most confidently preserved effects in the entire model – smaller in magnitude than the unconditional estimate of Section 3, but not remotely “explained away.” The class effect is real but largely indirect: Business passengers are more satisfied largely *because* they enjoy better seat comfort, entertainment and service, so in Section 3 the Class dummy was partly standing in for those ratings simply because they were not yet in the model.

Personal Travel gets stronger, not weaker. Section 3 found the type-of-travel effect nearly explained away by *Class*. Under the horseshoe, once the service ratings control for the *quality* of the trip experience, *Personal Travel* re-emerges with

a sizeable negative coefficient and a credible interval that lies below zero, although only just ($[-1.03, -0.002]$). The two findings are not contradictory: they show that the earlier attenuation was specifically about *Class* soaking up part of the travel-type signal, not about travel type being unimportant.

A cautionary sign on time convenience. More tentatively, *Departure/Arrival time convenient* has a *negative* posterior mean with a credible interval excluding zero – higher ratings associated with *lower* satisfaction, which is counter-intuitive. Recall from Section 1.3 that this item has one of the largest shares of structural zeros and sits in its own correlated cluster with *Seat comfort*, *Food and drink* and *Gate location*; its zeros plausibly encode “not applicable” rather than genuine dissatisfaction. We therefore read this coefficient as a candidate symptom of that ambiguity rather than a reliable substantive finding, and revisit it directly in the structural-zero robustness check of Section 5.3.

Loyalty dominates throughout. Finally, *disloyal Customer* remains, by a wide margin, the single strongest and most confidently preserved effect in the entire model – consistent across the exploratory analysis and both models of Sections 3 and 4, regardless of how many other covariates are present.

5. Prior Sensitivity and Structural Robustness

This section stress-tests the conclusions of Section 3 (vague-prior logit) and Section 4 (horseshoe) against the modelling choices we made along the way. Each of the four checks below opens with the question it answers. The chapter then closes by comparing the two fitted models directly via WAIC and PSIS-LOO [1], which formalises the choice of carrying the horseshoe model forward to Section 6.

5.1 Logit model prior sensitivity

Does the vague model’s posterior depend on the prior variance we picked? We refit Section 3’s model (vague priors, dummy Class encoding) four times, varying only the prior variance on every β_j :

$$\sigma_\beta^2 \in \{1, 10, 100, 10000\}.$$

Figure 4 overlays the posterior mean and 95% credible interval of each coefficient across the four prior variances.

Table 5 reports, for each coefficient, the range (max – min) of its posterior mean across the four prior variances.

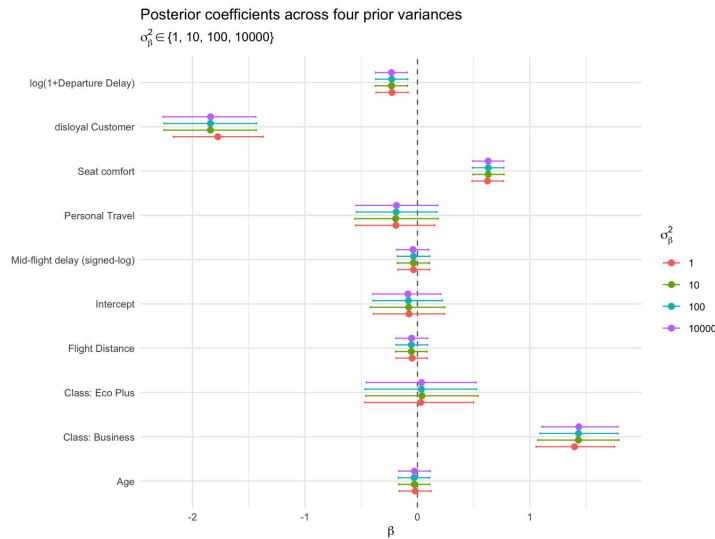


Table 5. Range of posterior means across the four prior variances (rows are ordered by decreasing range in the notebook).

Term	Range
Intercept	0.010
Age	0.009
Seat comfort	0.008
Flight Distance	0.006
Class: Eco Plus	0.012
Class: Business	0.038
log(1+Departure Delay)	0.005
Personal Travel	0.008
disloyal Customer	0.066
Mid-flight delay (signed-log)	0.004

Figure 4. Posterior coefficients of the vague-prior logit model across four prior variances.

The largest movement across four orders of magnitude of prior variance is only 0.066 on the log-odds scale, for a coefficient whose posterior mean is itself on the order of 1–2. At $n \approx 1200$ the likelihood therefore dominates all of the tested priors.

5.2 Horseshoe prior sensitivity: does the global scale matter?

The horseshoe’s shrinkage behaviour is governed by the scale s of the half-Cauchy prior placed on the **global** scale τ :

$$\tau \sim \mathcal{C}^+(0, s).$$

Small s pulls every coefficient hard towards zero (more aggressive sparsity); large s relaxes the global shrinkage. This is the parameter whose choice could plausibly change *which* covariates the model keeps, so it is the one worth testing.

We refit the Section 4 model three times, varying only this global scale, $s \in \{0.1, 1, 10\}$ ($s = 1$ is the Section 4 default), while the local scales λ_j keep the standard $\mathcal{C}^+(0, 1)$. For each fit it stores the posterior mean and 95% credible interval of every coefficient, and the per-predictor shrinkage weight

$$\kappa_j = \frac{1}{1 + \lambda_j^2}$$

(near 0 = signal, near 1 = shrunk, exactly as in Section 4).

5.2.1 Coefficients across global scales

Figure 5 overlays the posterior coefficients under the three global scales, and Table 6 lists the eight predictors whose posterior mean moves most across them.

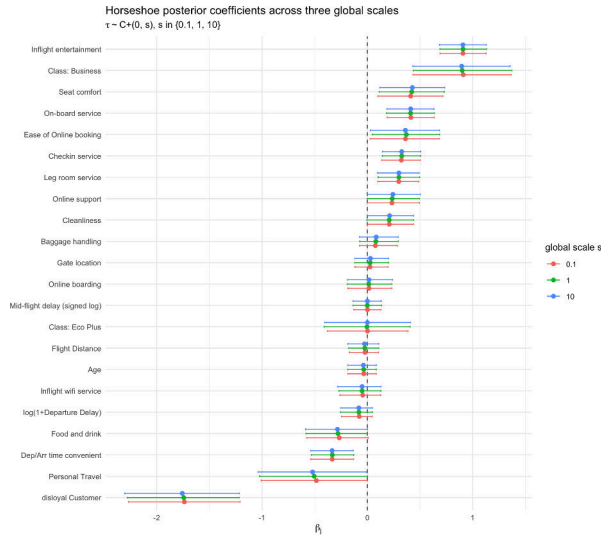


Table 6. Largest movement of the posterior mean across the three global scales (top eight predictors).

Term	Range
Personal Travel	0.036
disloyal Customer	0.020
Food and drink	0.017
Seat comfort	0.016
Class: Business	0.015
Ease of Online booking	0.009
Online support	0.009
Baggage handling	0.008

Figure 5. Horseshoe posterior coefficients across three global scales, ordered by shrinkage weight at the default scale.

5.2.2 Shrinkage weights and signal/noise stability

Turning each κ_j into a verdict ($\kappa_j < 0.5 \Rightarrow$ signal) lets us ask the sharper question: does the global scale change *which* covariates the model keeps? Across a hundred-fold range of s , two of the 22 predictors change their classification: Online support and Cleanliness, both of which sit within 0.02 of the $\kappa = 0.5$ boundary at the default scale and so flip on the smallest movement. The other twenty keep the same verdict throughout. The eleven the model calls signal at $s = 1$ (disloyal Customer, Inflight entertainment, Class: Business, Personal Travel, Seat comfort, On-board service, Ease of Online booking, Dep/Arr time convenient, Checkin service, Leg room service and Food and drink) are the same eleven at $s = 0.1$ and at $s = 10$, apart from the two boundary cases.

The strong effects of Section 4 (disloyal Customer, Personal Travel, Seat comfort) stay strong and confidently non-zero regardless of s , and the largest shift in any posterior mean is 0.036 on the log-odds scale. Only the coefficients already sitting near the boundary drift, which is exactly where sensitivity to the global scale is expected and where we drew no firm conclusions in the first place. The headline reading of Section 4 is therefore not an artefact of the particular half-Cauchy scale we chose. The full κ_j values at each scale, and the verdict table they produce, are in Appendix D.

5.3 Structural-zero robustness check

Does the counter-intuitive Departure/Arrival time convenient coefficient survive dropping the ambiguous structural-zero rows? Section 4 speculated that its counter-intuitive *negative* coefficient might be an artefact of the structural zeros flagged in Section 1.3 (zeros that plausibly mean “not applicable” rather than genuine dissatisfaction). We test that directly by refitting the full horseshoe model with those rows excluded: 110 rows are dropped, leaving 1103.

Table 7 reports the refit, and Figure 6 compares the “seat/catering/ground” cluster with and without the ambiguous rows. **The hypothesis does not survive the check.** Departure/Arrival time convenient’s negative coefficient does not shrink towards zero when the ambiguous rows are removed; it gets *more* negative (from -0.334 to -0.506), while its credible interval widens slightly, as expected with 110 fewer rows. Whatever drives this effect, it is not simply the structural-zero ambiguity we suspected in Section 4; we report this as an open, unresolved finding rather than force a tidier story onto it. On the other hand, the other two members of that same correlated cluster move in *opposite* directions once the

Table 7. Horseshoe refit excluding structural-zero rows: the ten predictors the horseshoe keeps as signal ($\kappa_j < 0.5$), plus **Food and drink**, discussed below. The other eleven move by less and are omitted.

Term	β mean	β lo	β hi	κ_j
disloyal Customer	−2.298	−2.998	−1.621	0.061
Inflight entertainment	+1.099	+0.844	+1.376	0.164
Seat comfort	+0.869	+0.550	+1.192	0.215
Class: Business	+0.936	+0.378	+1.518	0.219
Personal Travel	−0.678	−1.256	−0.037	0.308
On-board service	+0.526	+0.240	+0.809	0.343
Dep/Arr time convenient	−0.506	−0.780	−0.240	0.361
Ease of Online booking	+0.498	+0.081	+0.872	0.369
Checkin service	+0.376	+0.156	+0.592	0.429
Leg room service	+0.294	+0.050	+0.524	0.492
Food and drink	+0.025	−0.215	+0.299	0.732

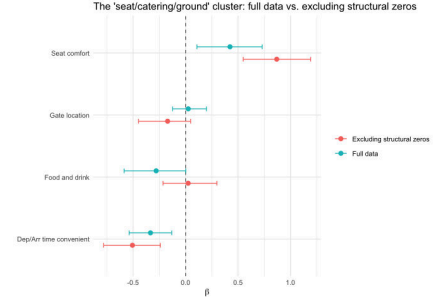


Figure 6. Coefficients of the correlated seat/catering/ground cluster, full data vs. excluding structural-zero rows.

ambiguous rows are excluded. **Seat comfort** strengthens substantially (κ drops from 0.370 to 0.215, β rises from +0.424 to +0.869), whereas **Food and drink** goes from a borderline signal ($\kappa = 0.486$, $\beta = -0.280$) to firmly shrunk ($\kappa = 0.732$, $\beta = +0.025$). The structural zeros therefore do not add uniform noise across the cluster; they affect its members differently.

5.4 Sample-size robustness

Do any of the conclusions depend on the size of the training set? The previous three checks varied the prior and which rows enter the fit; this one varies how many. We refit the model of Section 3 on a stratified subsample of 15003 records, twelve times the 1213 used throughout. This leaves the coefficients we rely on unmoved (**Class: Business** by 0.005, **disloyal Customer** by 0.004, **Seat comfort** by 0.023) and contracts the intervals by a median factor of 3.54, in line with the 3.52 that $\sqrt{n}$ asymptotics predicts. Four weak effects (**Flight Distance**, **Age**, **Personal Travel**, **Mid-flight delay**) acquire intervals excluding zero, which is a limitation of the $n = 1213$ analysis rather than a finding about the airline.

5.5 Model comparison: WAIC and PSIS-LOO

Alongside the WAIC-based comparison of Section 3.2 (used there to choose the **Class** encoding), we compare the vague-prior logit of Section 3 and the horseshoe model of Section 4 via Pareto-smoothed importance sampling leave-one-out cross-validation (PSIS-LOO), computed from the same pointwise log-likelihood draws already collected for each fit.

Model	elpd_loo (SE)	p_{loo}	LOO-IC
Vague-prior logit	−686.7 (16.7)	10.4	1373.4
Horseshoe	−491.9 (21.2)	21.8	983.8

Table 8. PSIS-LOO comparison between the two models, on the $n = 1213$ training set. All Pareto k diagnostics are good ($k < 0.7$) for both models, so the PSIS-LOO estimates are reliable without further caveats.

The horseshoe attains a substantially higher expected log predictive density ($\Delta\text{elpd} = -194.8$, $\text{SE} = 17.5$): the difference is more than eleven standard errors, a decisive preference by any conventional threshold. This agrees with the WAIC comparison of Section 3.2 – indeed the logit’s LOO-IC (1373.4) matches its WAIC (1373.4) to the first decimal, confirming the two approximations agree here – and justifies carrying the horseshoe model forward as the model of record for Section 6.

6. Prediction and validation

Everything up to this point has been estimated on the 1213-customer training subsample. This section confronts the model with the 128274 held-out customers set aside in Section 1, which were never used for fitting, standardisation or variable selection. Following the conclusion of Section 5, the horseshoe model is the one carried forward here; the comparison against the vague-prior logit was already established via WAIC and PSIS-LOO in Section 5.5, while the corresponding comparison against BAS is reported in Appendix G.

6.1 From posterior draws to predicted probabilities

For a held-out customer with covariate row $\tilde{x}_i$, the Bayesian point prediction is the posterior mean of the success probability, obtained by pushing every retained draw through the inverse link and averaging:

$$\hat{p}_i = \frac{1}{S} \sum_{s=1}^S \text{logit}^{-1} \left(\beta_0^{(s)} + \tilde{x}_i^\top \beta^{(s)} + \beta_{\text{mid}}^{(s)} \text{midflight_delay}_i \right), \quad (4)$$

This averages the transformed draws rather than transforming the averaged ones: logit^{-1} is non-linear, so $\mathbb{E}[g(\theta)] \neq g(\mathbb{E}[\theta])$, and only the former propagates posterior uncertainty. Test covariates are standardised with the *training* means and standard deviations, so nothing from the held-out set enters the fit.

6.2 Discrimination and probabilistic accuracy

We report two complementary summaries. The area under the ROC curve (AUC) measures how well the model *ranks* customers and is invariant to the cutoff. The Brier score [1],

$$\text{BS} = \frac{1}{n} \sum_{i=1}^n (\hat{p}_i - y_i)^2, \quad (5)$$

is a strictly proper scoring rule, minimised only when the stated probabilities match the true ones, so it rewards predictions that are both well ranked and numerically honest. On the held-out set the horseshoe model attains an AUC of 0.900 and a Brier score of 0.126.

6.3 Classification and the choice of cutoff

Turning probabilities into hard classifications requires a cutoff. Table 9 contrasts the conventional 0.5 with the training base rate $\bar{y} = 0.546$, the natural alternative when the two classes are not perfectly balanced.

Table 9. Classification performance on the held-out set at two cutoffs.

Cutoff	Sensitivity	Specificity	Accuracy
$t = 0.5$	0.837	0.813	0.826
$t = \bar{y} = 0.546$	0.815	0.844	0.828

Moving to the base rate trades about two points of sensitivity for three of specificity and leaves accuracy essentially unchanged (0.826 against 0.828). That near-indifference is informative: the errors are not concentrated on one side, and accuracy alone is a blunt instrument for choosing an operating point. The right cutoff depends on what the two mistakes cost, which is Section 6.6.

6.4 Calibration

Neither AUC nor accuracy asks whether the predicted probabilities *mean* what they claim: among customers the model calls 70% likely to be satisfied, roughly 70% should be. This matters as much as discrimination here, since the posterior predictive probability is the object we report. Table 10 bins the held-out set into deciles of predicted probability against the observed rate.

Table 10. Calibration by decile of predicted probability.

Decile	Mean predicted	Observed rate	n
1	0.035	0.121	12828
2	0.100	0.126	12828
3	0.195	0.162	12828
4	0.326	0.260	12828
5	0.488	0.431	12827
6	0.657	0.637	12827
7	0.801	0.821	12827
8	0.896	0.936	12827
9	0.946	0.984	12827
10	0.977	0.997	12827

Calibration is good in the upper half: from the seventh decile up, predicted and observed agree closely (0.801 against 0.821, 0.896 against 0.936). In the middle the model is mildly over-confident about dissatisfaction, predicting 0.33 and 0.49 in deciles four and five where the observed rates are 0.26 and 0.43, and the lowest two deciles reverse that. So the model separates the classes well at the extremes, where most predictions live, and is least trustworthy around $\hat{p} \approx 0.3$ –0.6, where the evidence genuinely is ambiguous.

6.5 Posterior predictive check by subgroup

The checks above compare point predictions against outcomes. A posterior predictive check asks more: taking the fitted model seriously as a generative description and simulating whole replicated datasets from it, do those replications reproduce features of the real data the fit never targeted? For each posterior draw we simulate outcomes for the entire held-out set and recompute the satisfaction rate within each level of `Class`, `Customer Type` and `Type of Travel`, giving a posterior predictive distribution per subgroup to compare the observed rate against; Figure 7 shows the result.

The observed rates fall inside the bulk of the simulated distributions for every subgroup of all three variables, so the model is not systematically over- or under-predicting satisfaction for any class of customer on data it has never seen. The subgroup rates were never fitted directly, so reproducing them is evidence that the covariate effects were captured rather than merely tuned to the aggregate.

6.6 Choosing the cutoff by expected loss

Rather than pick 0.5 or the ROC-optimal point by convention, we state what the two errors cost and minimise posterior expected loss. The airline flags a customer when $\hat{p}_i < t$ and follows up with a retention contact; write C_{miss} for the cost of missing a dissatisfied customer and C_{waste} for that of contacting a satisfied one. The expected loss at cutoff t is

$$\mathcal{L}(t) = C_{\text{miss}} \cdot \mathbb{P}(\hat{p} \geq t, y = 0) + C_{\text{waste}} \cdot \mathbb{P}(\hat{p} < t, y = 1), \quad (6)$$

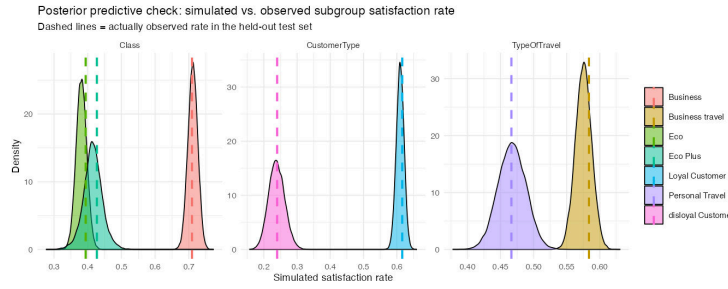


Figure 7. Posterior predictive check by subgroup. Densities are the simulated replicate satisfaction rates; the vertical line is the observed rate.

Table 11. Cutoff minimising expected loss, by assumed cost ratio $C_{\text{miss}} : C_{\text{waste}}$.

Cost ratio	Optimal cutoff t^*	Expected loss
1 : 1 (equal cost)	0.55	0.172
3 : 1	0.75	0.252
5 : 1	0.80	0.288

estimated on the held-out set. Only the *ratio* of the costs matters, so the airline need not price either error, just say how many wasted contacts it would trade for one retained customer. Table 11 gives the minimising cutoff at three such ratios.

Under equal costs the optimum is $t^* = 0.55$, essentially the base rate and close to the conventional 0.5, a useful sanity check on Equation (6). As missing a dissatisfied customer grows more expensive the cutoff rises, to 0.75 at 3 : 1 and 0.80 at 5 : 1: the airline flags far more customers, accepting wasted contacts to catch the recoverable ones. This is where the Bayesian machinery pays off operationally. Because the model returns a calibrated distribution rather than a label, the same fit supports any cost structure the business specifies, and the cutoff follows from the loss function instead of convention.

7. Conclusions

We modelled airline customer satisfaction with a Bernoulli-logit likelihood, first on the reduced covariate set suggested by the brief and then on the full set under a horseshoe shrinkage prior, with BAS model averaging as an independent cross-check. All posteriors were sampled by MCMC in JAGS.

7.1 Findings

Three results stand out, and survive every specification:

1. **Customer loyalty** is the strongest effect by a wide margin and the least shrunk coefficient at every global scale tested ($\kappa_j = 0.07$).
2. **Inflight entertainment** and **Business** class follow, essentially tied ($\kappa_j = 0.17$ and 0.18). Entertainment is the service item the prior preserves most clearly.
3. **Inflight wifi service** is heavily shrunk ($\kappa_j = 0.73$) despite correlating at **0.67** with **Online boarding** inside the same booking cluster. That asymmetry is what the horseshoe was chosen for.

These conclusions are not artefacts of our modelling choices. The vague-prior posterior moves by at most 0.066 across four orders of magnitude of prior variance; two predictors of twenty-two change their signal/noise verdict across a hundred-fold range of the global scale, and both sit within 0.02 of the $\kappa = 0.5$ boundary; and a training set twelve times larger moves the coefficients we report by at most 0.023. Out of sample, on 128274 unseen customers, the model reaches an AUC of **0.900**, classifies **82.8%** correctly at the base-rate cutoff, and reproduces the observed satisfaction rate in every subgroup.

7.2 Open questions and limitations

The Departure/Arrival time convenient coefficient. It is negative, its interval excludes zero, and it stayed negative when we refit without the structural-zero rows, which is the direction the ambiguity would have had to run to explain it. Dropping those rows strengthened the effect rather than weakening it, from -0.334 to -0.506 . We report it as an open question rather than a finding: the check we designed to explain it away did the opposite, and nothing else in the analysis accounts for it.

The structural zeros. Several service-quality items contain zeros that plausibly mean “not applicable” rather than genuine dissatisfaction. They were flagged and tested by refitting without them, but not modelled: a zero-inflated likelihood would treat the two kinds of zero as what they are, and would not require discarding 110 rows to ask the question.

Three local scales did not converge. Three of the horseshoe’s λ_j did not reach the conventional 1.1 threshold (Appendix C), the worst at 1.21. Tripling the chain length left them unchanged, so this is weak identification rather than insufficient sampling: for a covariate the data barely constrains, the amount of shrinkage applied to it is itself barely constrained. The coefficients those scales act on did converge, and Section 5.2 shows the signal/noise verdicts survive a hundred-fold change in the global scale, so the findings above stand. The precise shrinkage weight of three weak covariates, though, should not be read to three decimals.

A. Data augmentation

A.1 Decorrelating departure and arrival delays

Inspecting the correlation matrix (Fig. 1a), we notice an extremely high correlation between `Departure delay` and `Arrival delay`. This value makes the two variables almost duplicates of one another. In order to remove this redundancy while retaining as much information as possible, we decided to replace `Arrival delay` with a quantity that captures the difference between the two.

We considered two options:

- using the plain difference between the two variables;
- fitting a simple linear regression model with `Arrival delay` as the target and `Departure delay` as the covariate, and taking the residual of each observation as the new covariate.

The second model is more general and, in fact, encompasses the first one; however, the first is more interpretable and carries an immediate meaning, such as “mid-flight delay” or “time gained during the flight”. The deciding point was whether the second option differed enough from the first to justify its higher complexity.

The regression model that reproduces the plain difference is the one with slope equal to 1 and intercept equal to 0. Fitting the regression model on the full dataset yields an intercept of 0.76 and a slope of 0.97; we therefore decided to opt for the simple difference between the arrival and departure delays.

A.2 Reducing covariates skewness

From the boxplots in Fig.8 we can see how covariates `Flight Distance`, `Departure Delay` and `Mid-flight delay` are extremely right-skewed. Since covariate skewness might slow down MCMC convergence and are prone to present outliers and leverage points, we decided to transform them and check whether the transformation is beneficial.

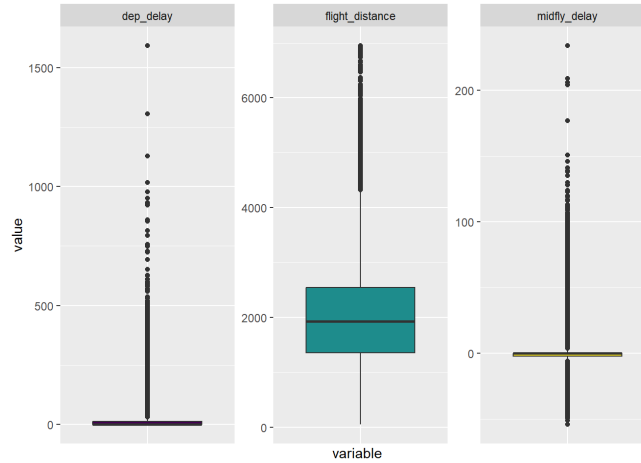


Figure 8. Boxplots of skewed covariates.

Since both `Flight distance` and `Departure delay` are non-negative we transform them using $\ln(1+x)$; instead for `Mid-flight delay` we use the *signed-log transformation* $\text{sign}(x) \cdot \ln(1+|x|)$ due the presence of negative values.

variable	skewness
Flight Distance	0.466
$\log_{1p}(\text{Flight Distance})$	-1.599
Departure Delay	6.853
$\log_{1p}(\text{Departure Delay})$	0.919
Midflight Delay	2.941
$\text{signed-log}(\text{Midflight delay})$	0.234

Table 12. Skewness of raw and transformed variables

As we can see in Tab. 12 `Flight distance` is only mildly skewed and the transformation actually turn to be worse than the original variable, therefore we kept it unchanged. Instead for what concerns `Departure delay` and `Mid-flight delay` we see higher level of skewness and the transformation improves this aspect; therefore we decided to keep the log-transformed variables.

B. Low-informative logit model MCMC convergence

The following section reports the plots and indices used to assess the convergence of our MCMC used to estimate the simple logit model shown in Section 3. In

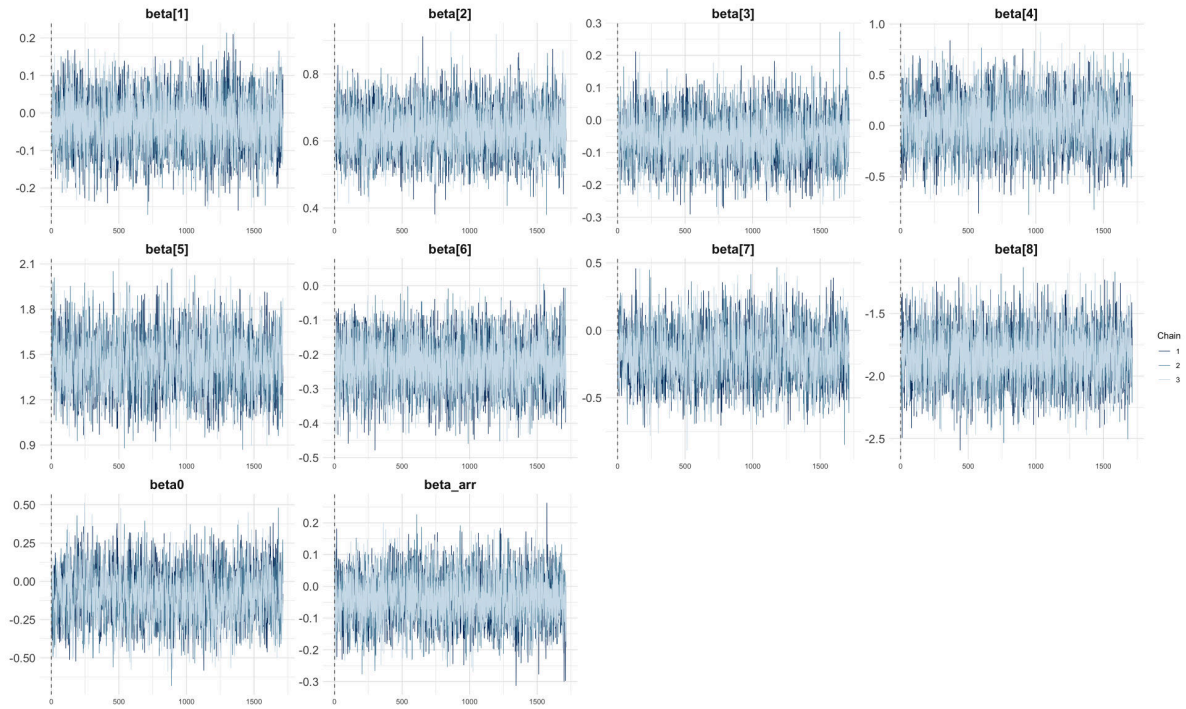


Figure 9. MCMC traceplot of the baseline logit model.

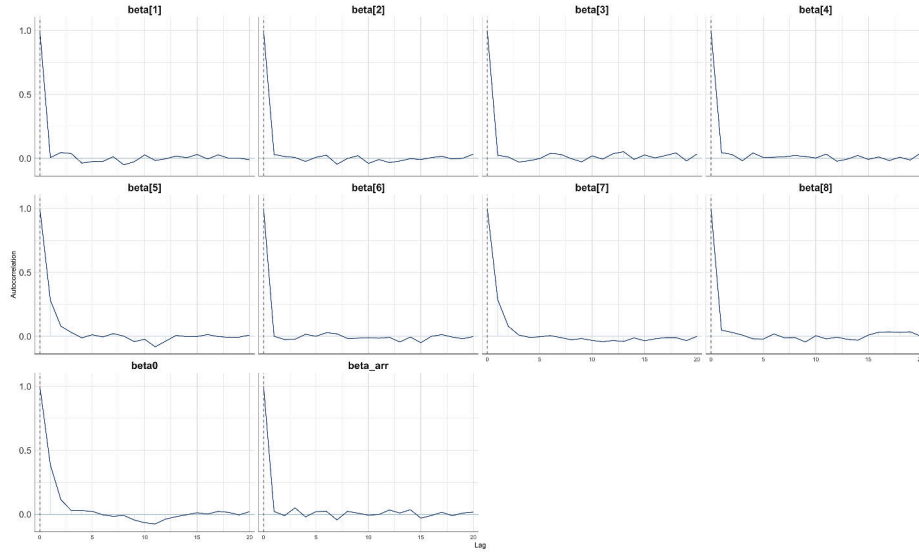


Figure 10. MCMC ACF plots of the baseline logit model (1st chain).

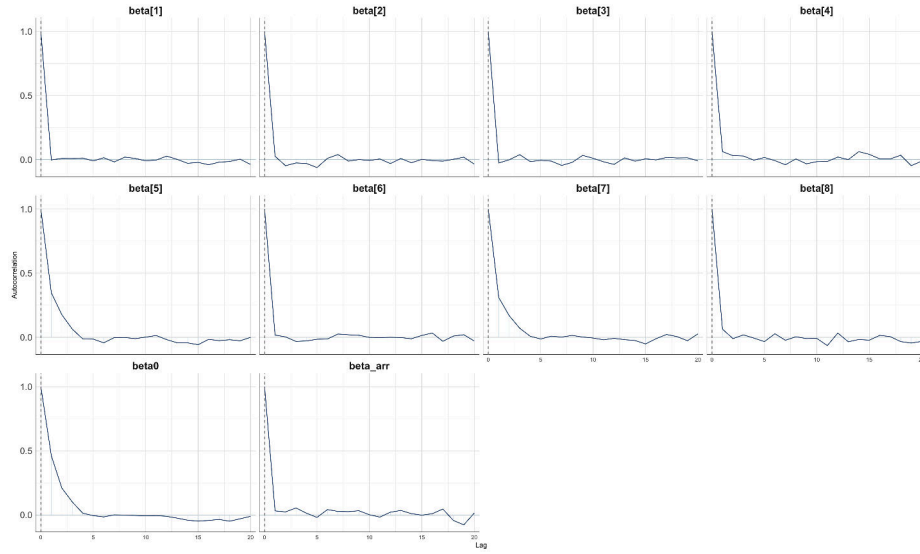


Figure 11. MCMC ACF plots of the baseline logit model (2nd chain).

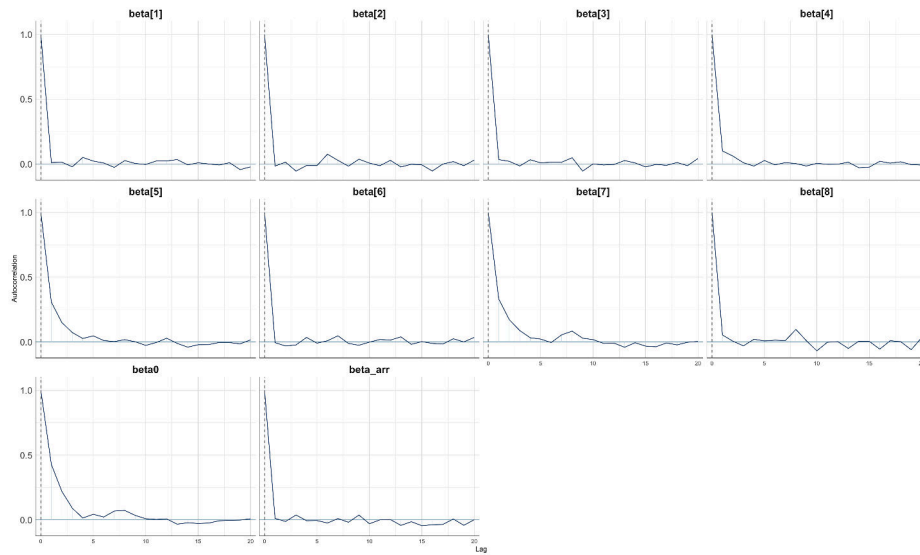


Figure 12. MCMC ACF plots of the baseline logit model (3rd chain).

Table 13. Gelman–Rubin potential scale reduction factors.

Parameter	Point est.	Upper C.I.
β_1	1	1.00
β_2	1	1.00
β_3	1	1.00
β_4	1	1.01
β_5	1	1.00
β_6	1	1.01
β_7	1	1.00
β_8	1	1.01
β_0	1	1.01
β_{mid}	1	1.00
Multivariate PSRF: 1		

Table 14. Effective sample sizes.

Parameter	ESS
β_1	5142
β_2	5370
β_3	5142
β_4	4147
β_5	2456
β_6	5142
β_7	2447
β_8	5169
β_0	2057
β_{mid}	4476

C. Horseshoe model MCMC convergence

This appendix collects the diagnostics behind the convergence claims made in Section 4 for the horseshoe model. Three chains, 30000 retained draws each after a burn-in of 1000 and thinning of 3. Diagnostics are reported separately for the two blocks of parameters: the regression coefficients $\{\beta_j, \beta_0, \beta_{\text{mid}}\}$ and the scale parameters $\{\lambda_j, \lambda_{\text{mid}}, \tau\}$, the latter being the harder to sample.

Trace plots

Figures 3a and 3b in Section 4 show the first group of each block. The remaining groups follow.

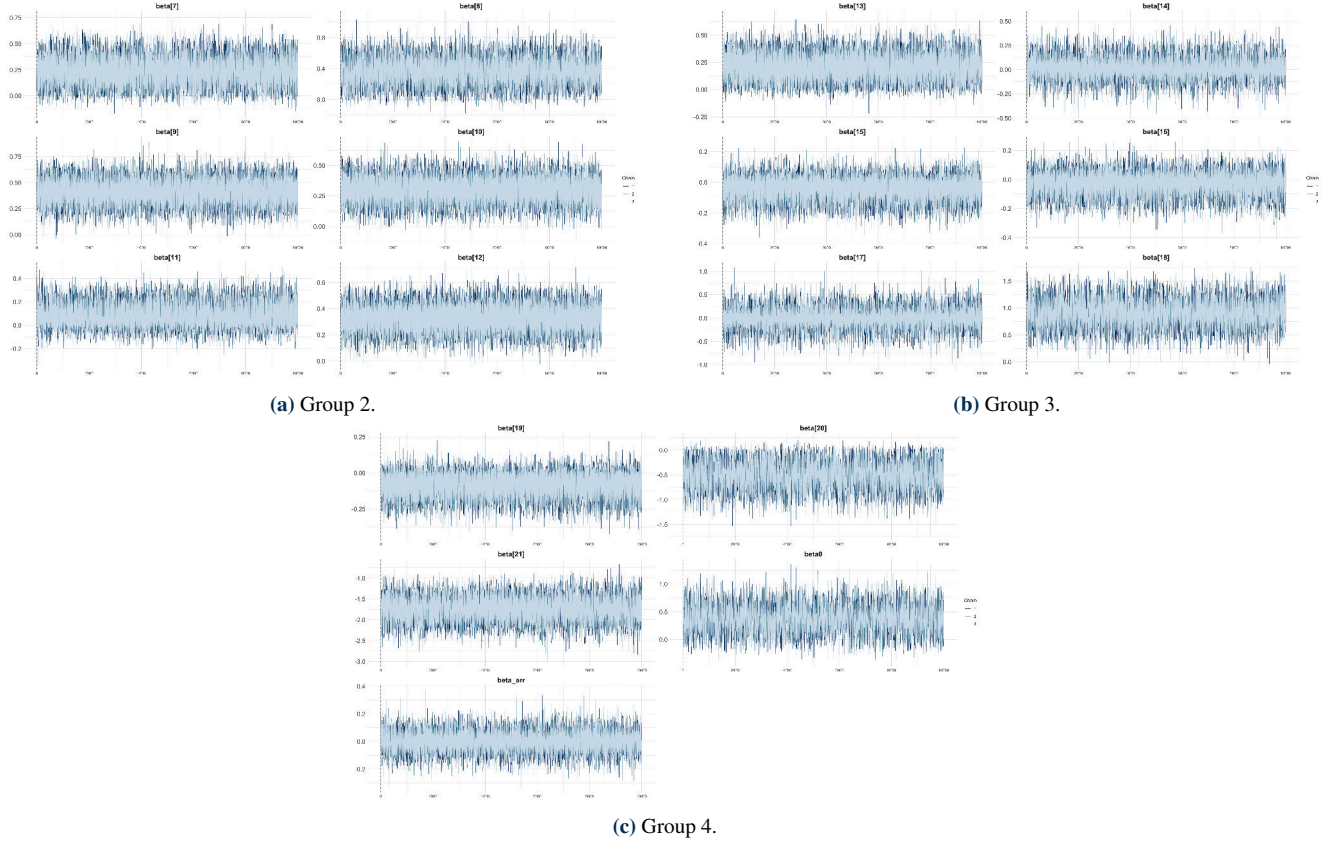


Figure 13. Trace plots for the remaining regression coefficients β_j , three chains.

Autocorrelation

Autocorrelation functions for every parameter, one set per chain. The coefficients decay within a few lags; the local scales decay more slowly, which is the expected behaviour for a horseshoe and the reason for the thinning.

Gelman–Rubin and effective sample sizes

Every coefficient has a potential scale reduction factor of 1.00 and an effective sample size in the thousands, the lowest being the intercept at 3412. The scale parameters are less well behaved, as expected: their effective sample sizes run from 1388 to 4882, and three local scales sit at or above the conventional 1.1 threshold — λ_{10} (*Leg room service*) at 1.21 with an upper limit of 1.31, λ_3 (*Food and drink*) at 1.10, and λ_{14} (*Online boarding*) at 1.10. We report this rather than smooth it over. The quantity these feed into is the shrinkage weight $\kappa_j = 1/(1 + \lambda_j^2)$, which is a bounded, monotone transformation of λ_j , and the corresponding coefficients β_3 , β_{10} and β_{14} have themselves converged cleanly (PSRF 1.00, ESS 7253, 19144 and 12963). The conclusions of Section 4 rest on the coefficients and on the broad signal/noise separation, not on the third decimal of any single κ_j , and Section 5.2 shows the verdicts are stable across a hundred-fold change in the global scale. The global scale τ itself mixes well (PSRF 1.00, ESS 4515).

D. Horseshoe shrinkage weights

The shrinkage weight $\kappa_j = 1/(1 + \lambda_j^2)$ of Eq. (3) measures how hard the prior pulls each coefficient towards zero: near 0 the signal is preserved, near 1 the coefficient is treated as redundant. Figure 21 shows the posterior distribution of each κ_j at the default global scale, supporting the relevance ranking of Section 4. Table 17 gives the values and the resulting signal/noise verdicts at each of the three global scales tested in Section 5.2.

Figure 3c in Section 4 shows the first group of marginal posterior densities. The remaining three follow.

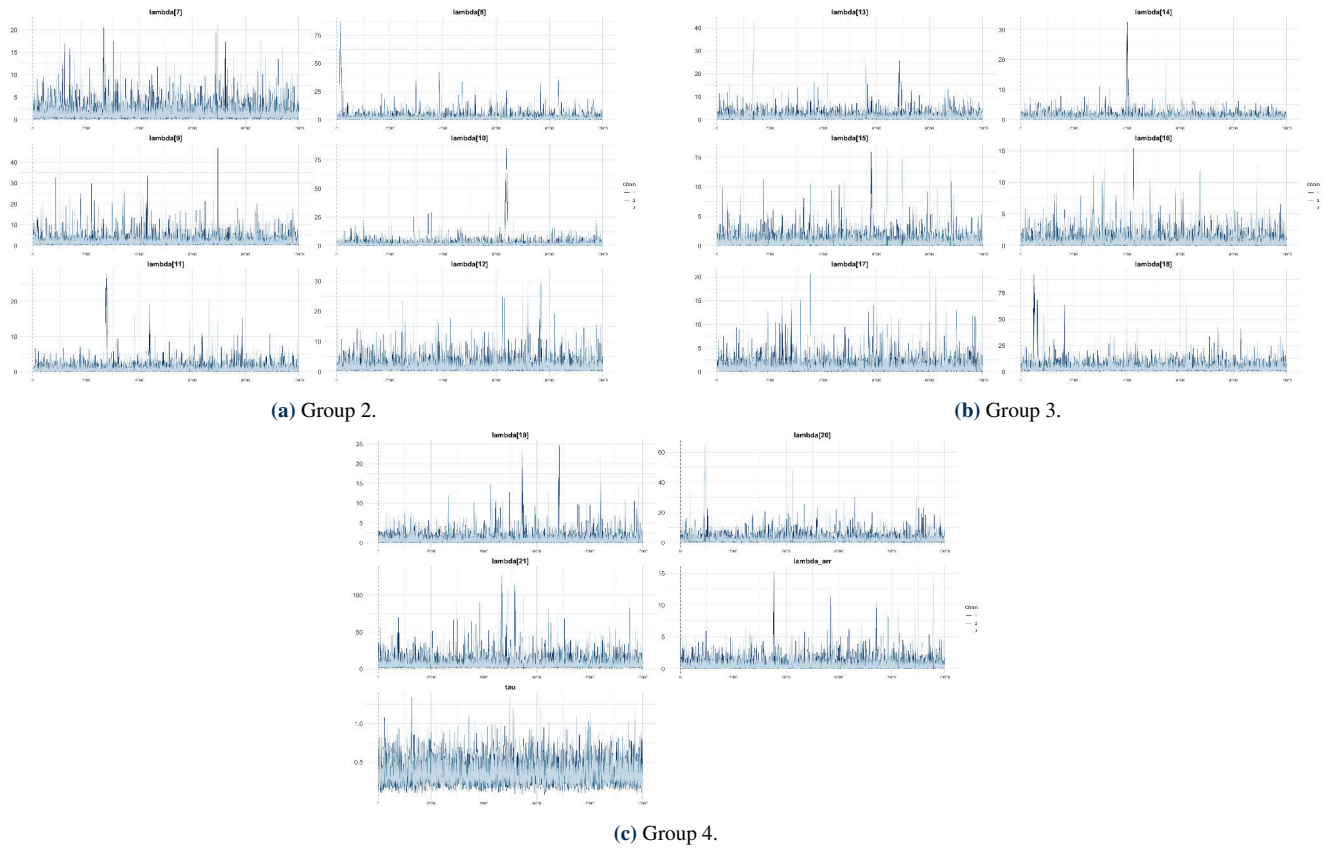


Figure 14. Trace plots for the remaining scale parameters λ_j , three chains.

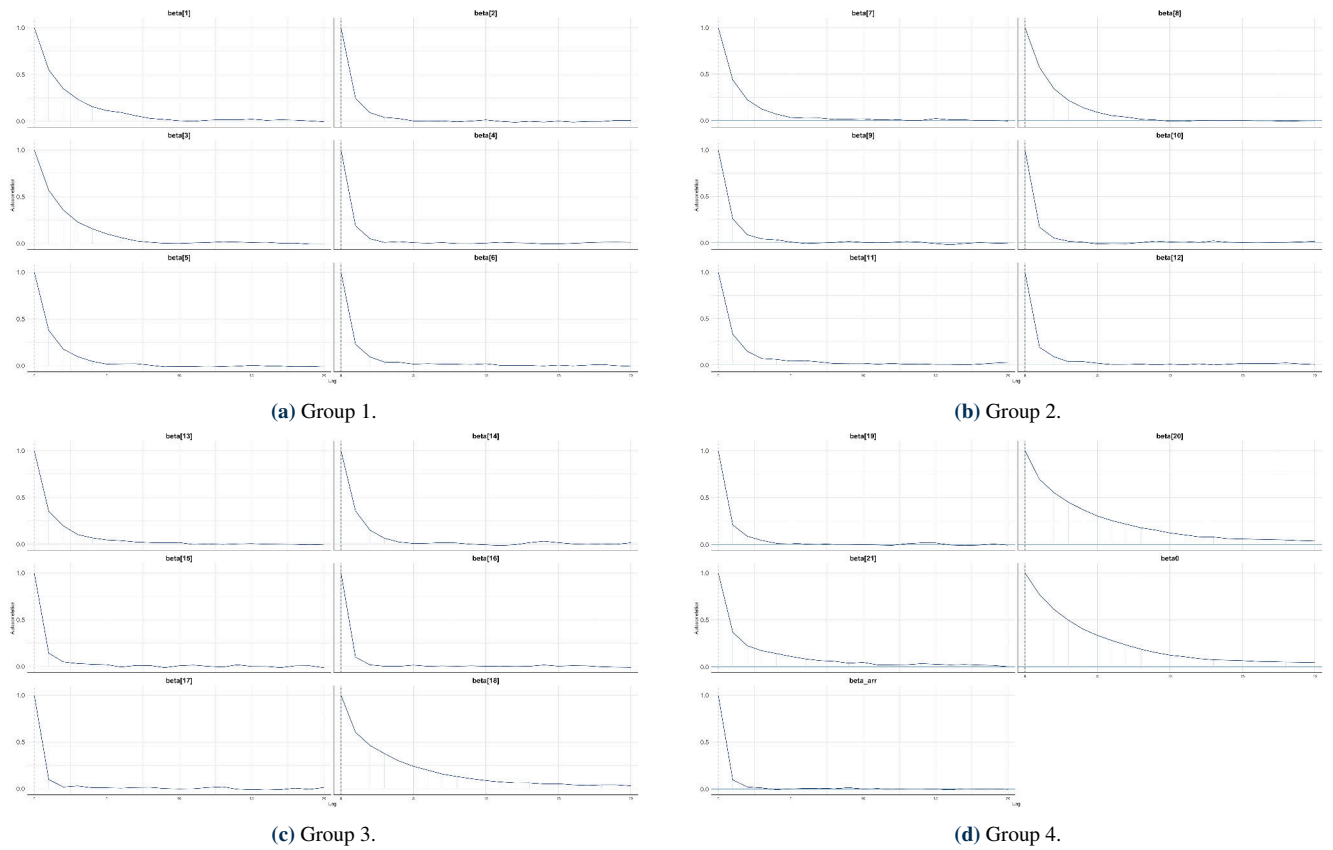


Figure 15. Autocorrelation of the coefficients β_j , chain 1.

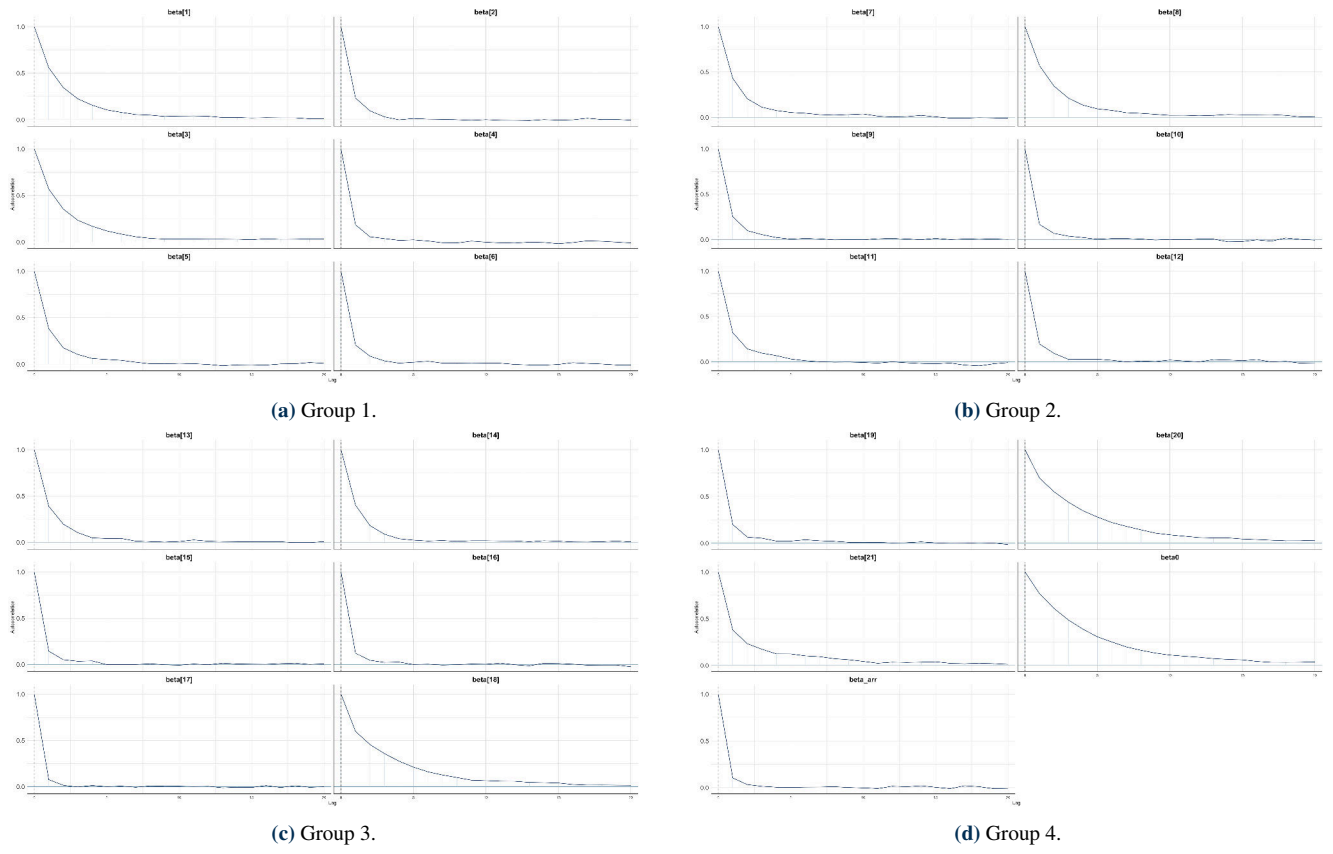


Figure 16. Autocorrelation of the coefficients β_j , chain 2.

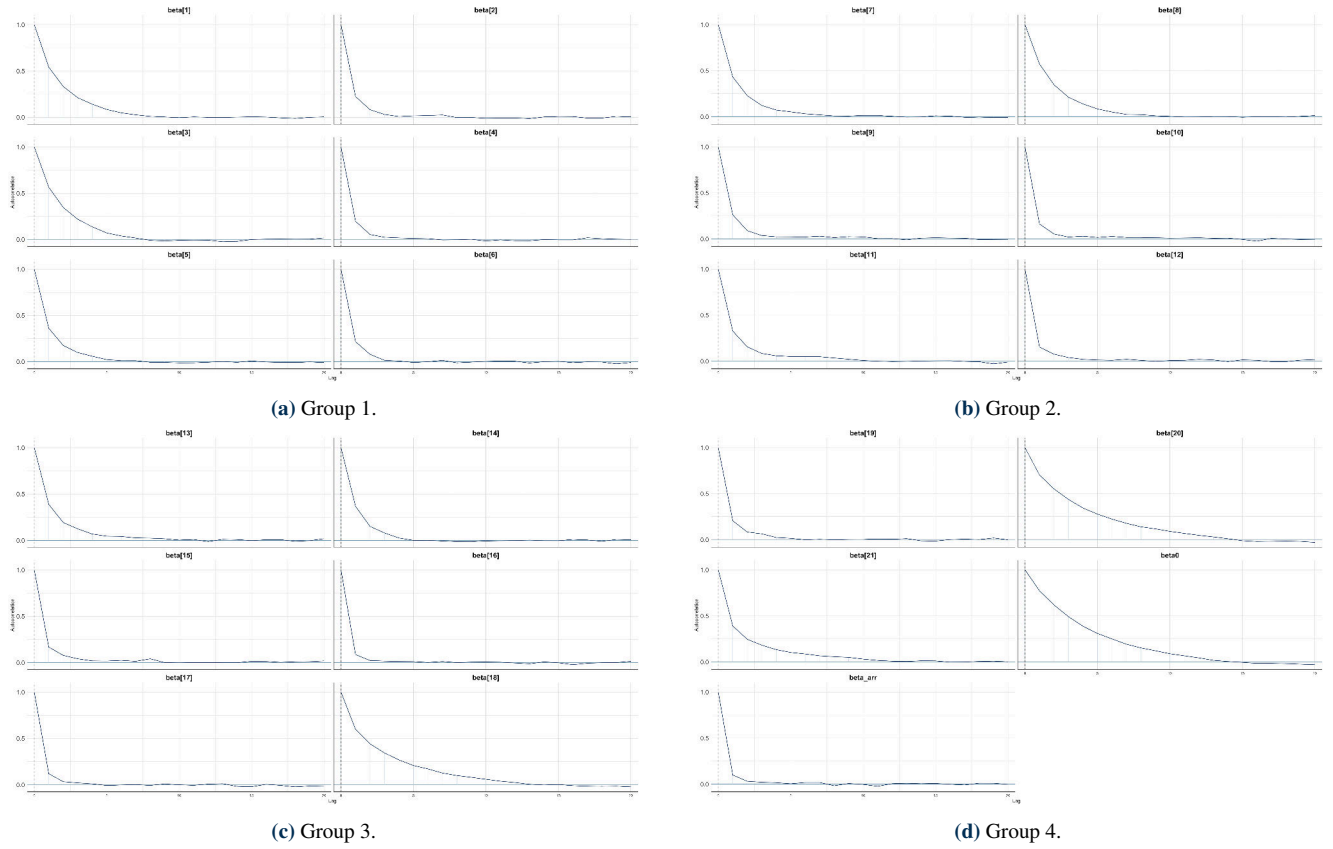


Figure 17. Autocorrelation of the coefficients β_j , chain 3.

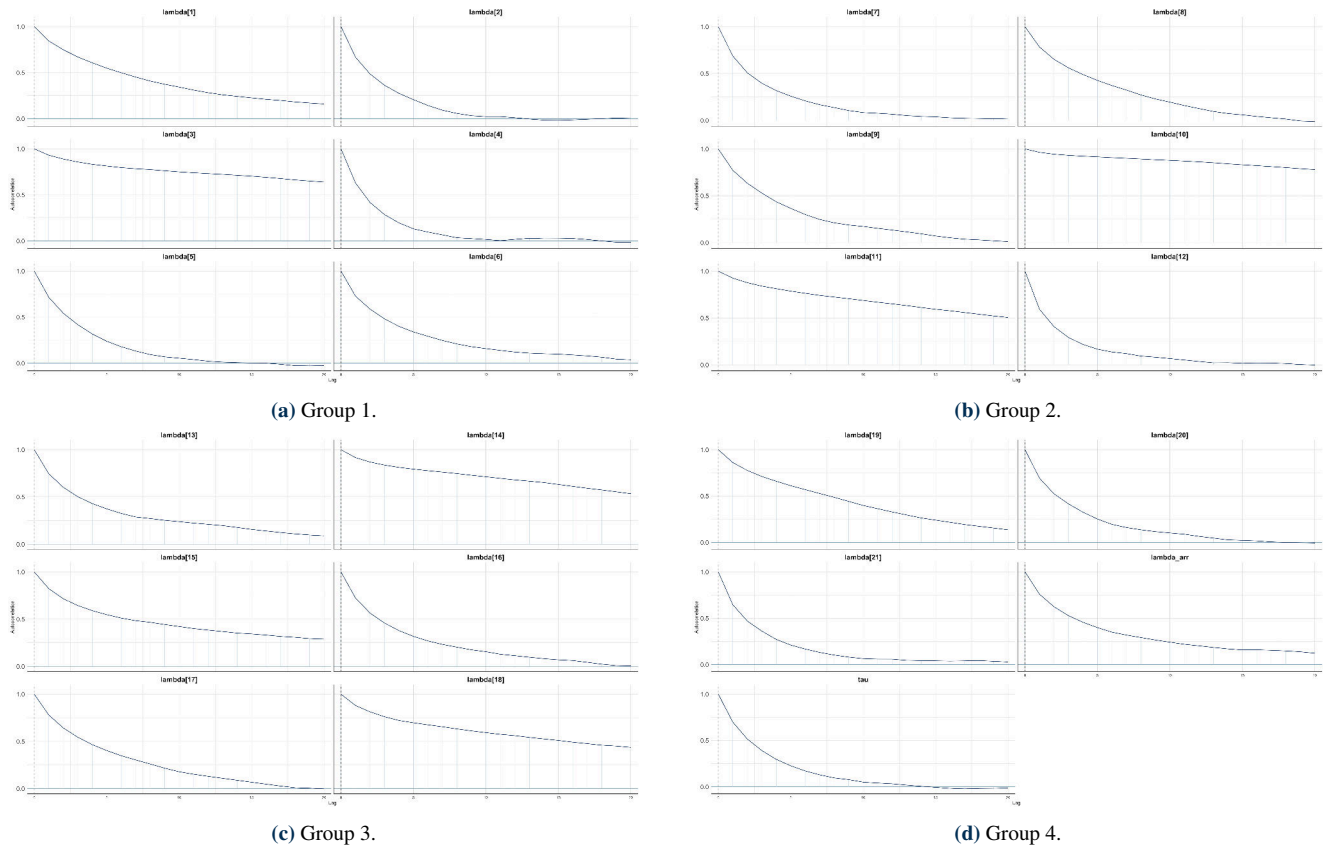


Figure 18. Autocorrelation of the scales λ_j , chain 1.

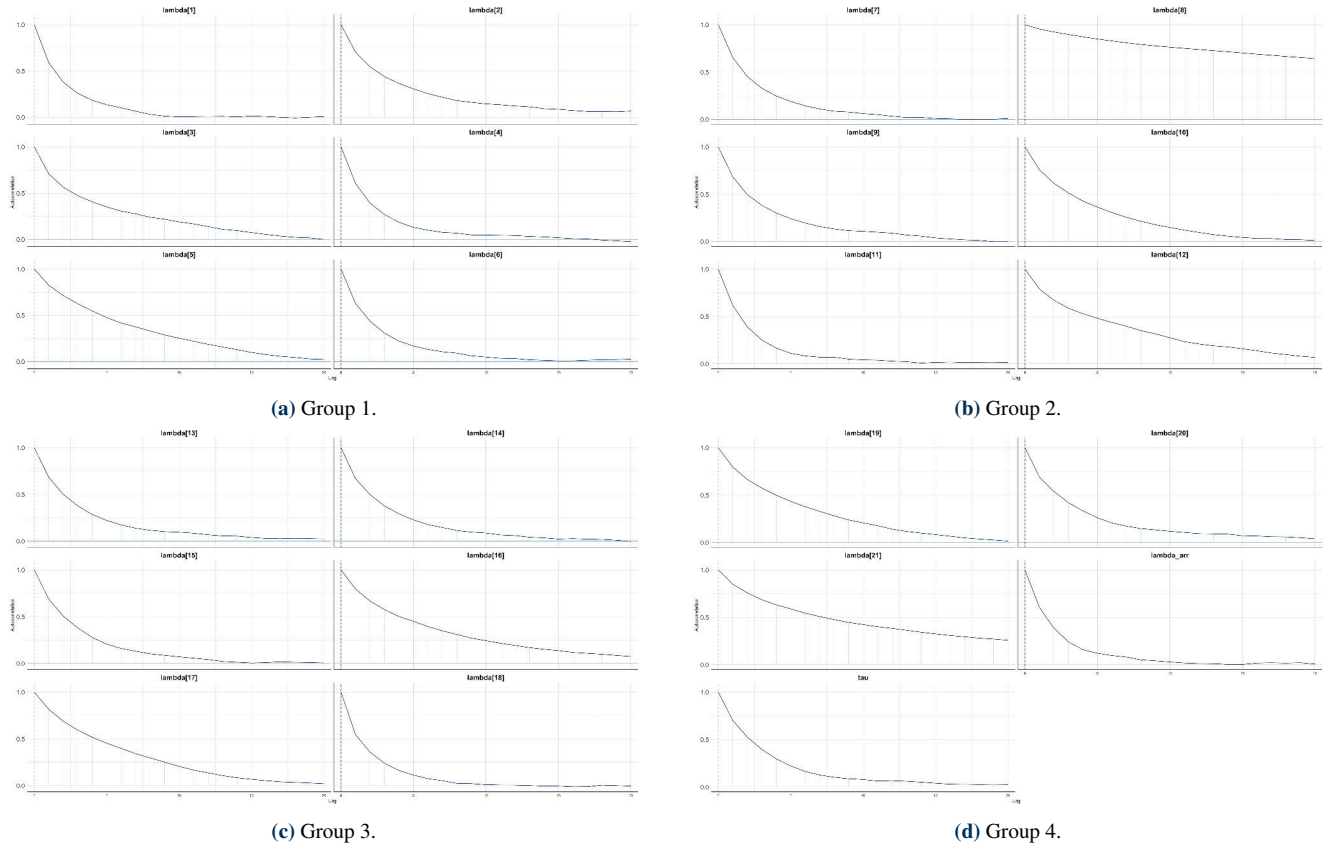


Figure 19. Autocorrelation of the scales λ_j , chain 2.

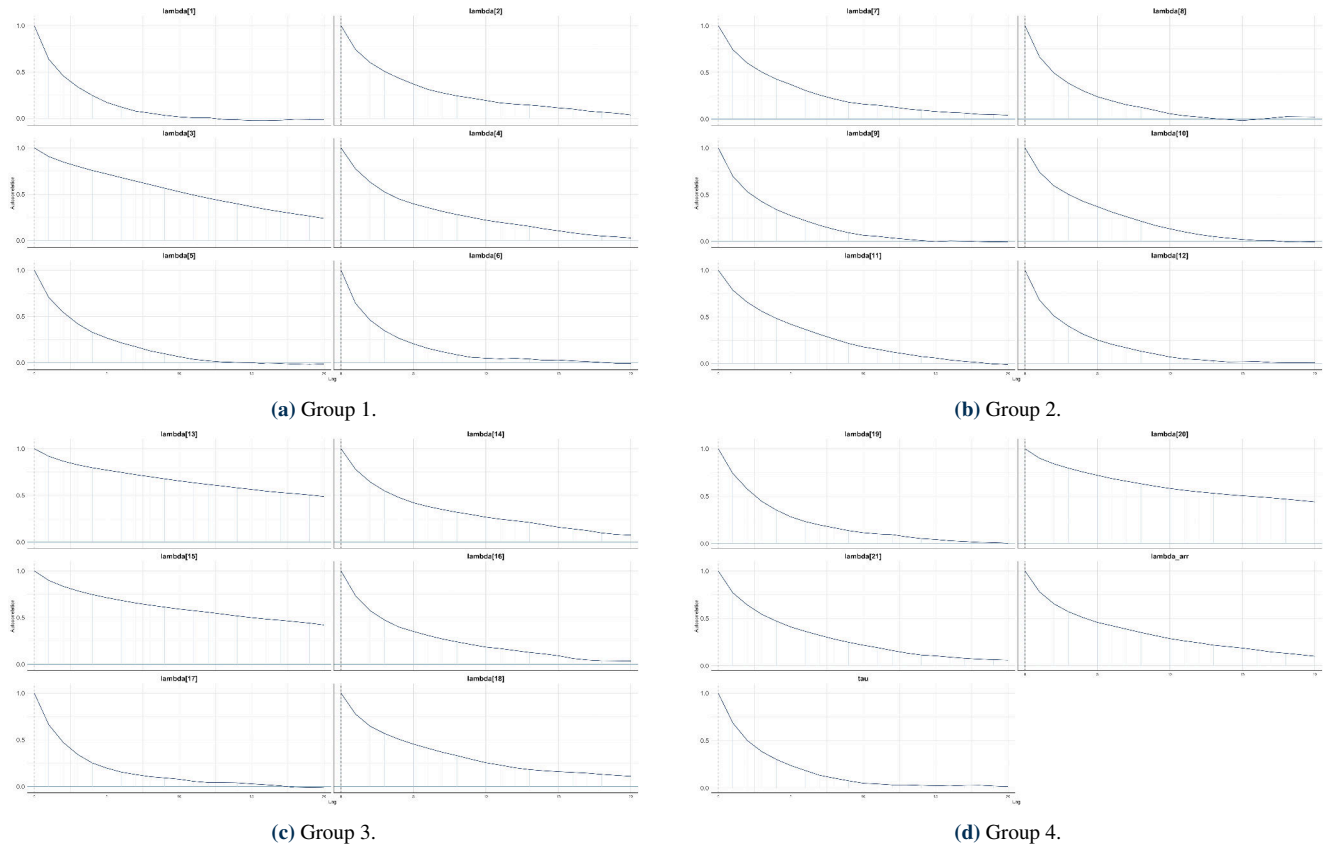


Figure 20. Autocorrelation of the scales λ_j , chain 3.

Table 15. Regression coefficients: potential scale reduction factors and effective sample sizes.

Parameter	Point est.	Upper C.I.	ESS
β_1 (Seat comfort)	1.00	1.00	7252
β_2 (Dep/Arr time convenient)	1.00	1.00	17305
β_3 (Food and drink)	1.00	1.00	7253
β_4 (Gate location)	1.00	1.00	19441
β_5 (Inflight wifi service)	1.00	1.00	11754
β_6 (Inflight entertainment)	1.00	1.00	16866
β_7 (Online support)	1.00	1.00	10699
β_8 (Ease of Online booking)	1.00	1.00	7660
β_9 (On-board service)	1.00	1.00	15288
β_{10} (Leg room service)	1.00	1.00	19144
β_{11} (Baggage handling)	1.00	1.00	11739
β_{12} (Checkin service)	1.00	1.00	17602
β_{13} (Cleanliness)	1.00	1.00	11220
β_{14} (Online boarding)	1.00	1.00	12963
β_{15} (Age)	1.00	1.00	18345
β_{16} (Flight Distance)	1.00	1.00	23187
β_{17} (Class: Eco Plus)	1.00	1.00	23539
β_{18} (Class: Business)	1.00	1.00	4767
β_{19} (log(1+Departure Delay))	1.00	1.00	16674
β_{20} (Personal Travel)	1.00	1.00	3813
β_{21} (disloyal Customer)	1.00	1.00	7952
β_0 (Intercept)	1.00	1.00	3412
β_{mid} (Mid-flight delay)	1.00	1.00	23618

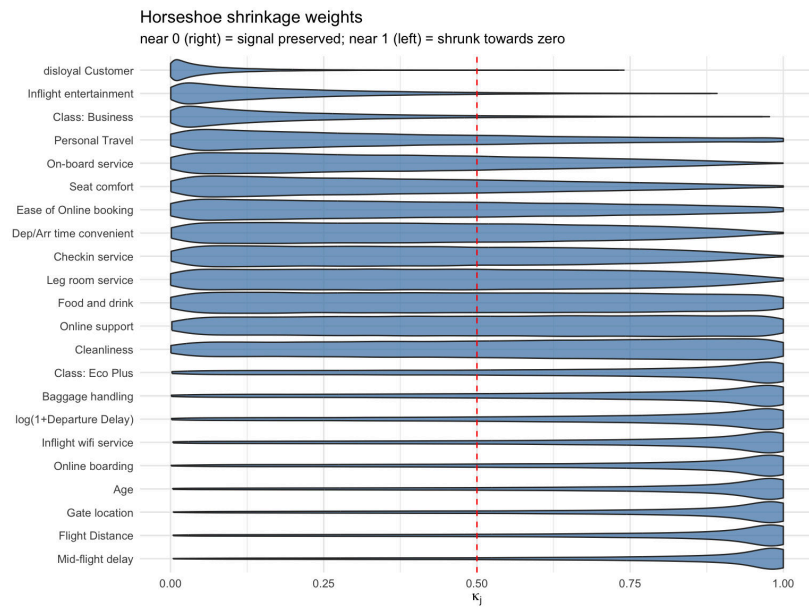


Figure 21. Posterior shrinkage weights κ_j under the horseshoe prior at the default global scale. Near 0 (right) the signal is preserved, near 1 (left) the coefficient is shrunk; the dashed line marks $\kappa_j = 0.5$.

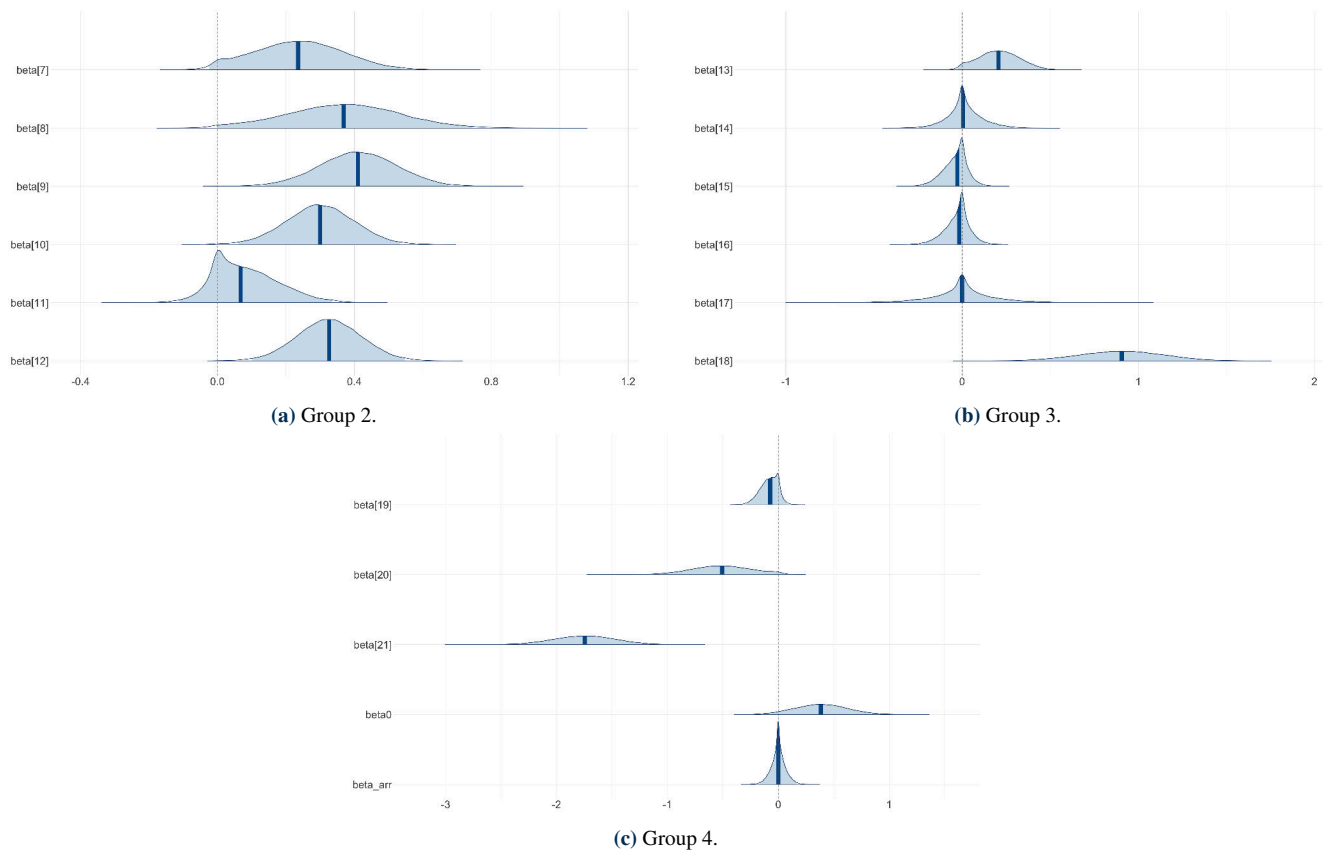


Figure 22. Marginal posterior densities for the remaining regression coefficients β_j under the horseshoe prior.

Table 16. Scale parameters: potential scale reduction factors and effective sample sizes.

Parameter	Point est.	Upper C.I.	ESS
λ_1 (Seat comfort)	1.01	1.01	4488
λ_2 (Dep/Arr time convenient)	1.02	1.02	3762
λ_3 (Food and drink)	1.10	1.12	1388
λ_4 (Gate location)	1.02	1.03	4882
λ_5 (Inflight wifi service)	1.03	1.04	3382
λ_6 (Inflight entertainment)	1.02	1.02	4499
λ_7 (Online support)	1.01	1.01	4077
λ_8 (Ease of Online booking)	1.02	1.03	2452
λ_9 (On-board service)	1.01	1.01	3662
λ_{10} (Leg room service)	1.21	1.31	2103
λ_{11} (Baggage handling)	1.03	1.04	3177
λ_{12} (Checkin service)	1.04	1.05	3822
λ_{13} (Cleanliness)	1.04	1.04	2497
λ_{14} (Online boarding)	1.10	1.11	2432
λ_{15} (Age)	1.08	1.11	2108
λ_{16} (Flight Distance)	1.00	1.00	2753
λ_{17} (Class: Eco Plus)	1.02	1.02	3341
λ_{18} (Class: Business)	1.03	1.03	3262
λ_{19} (log(1+Departure Delay))	1.05	1.06	2458
λ_{20} (Personal Travel)	1.01	1.02	2919
λ_{21} (disloyal Customer)	1.05	1.06	2755
λ_{mid} (Mid-flight delay)	1.03	1.04	3422
τ (global scale)	1.00	1.00	4515

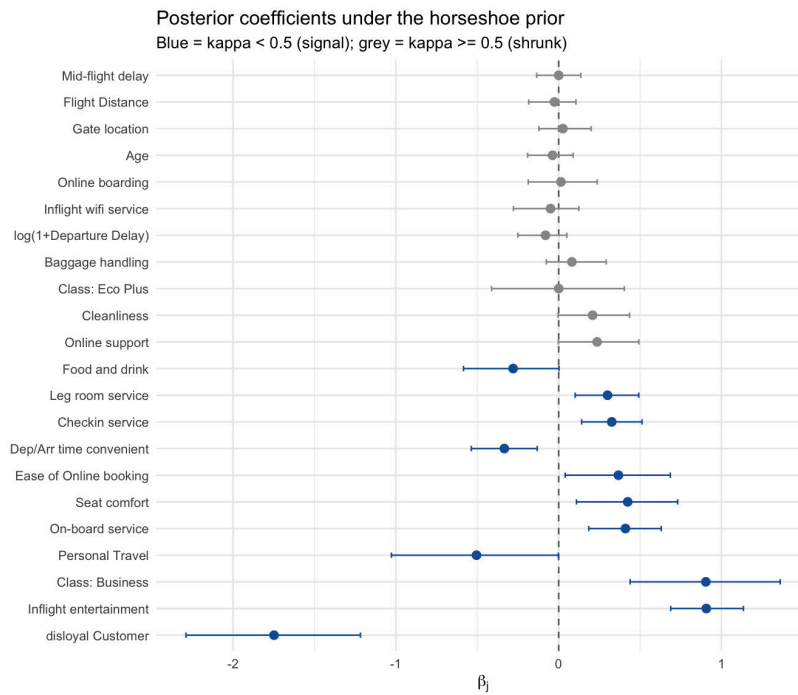
**Figure 23.** Posterior coefficients β_j under the horseshoe prior (means and 95% credible intervals). Blue marks the covariates identified as signal ($\kappa_j < 0.5$), grey the shrunk ones.

Table 17. Posterior shrinkage weight κ_j at each of the three global scales, ordered by κ_j at $s = 1$. Rows above the mid-rule are signal ($\kappa_j < 0.5$) at the default scale; ✓ marks the predictors whose signal/noise verdict is the same at all three scales.

Term	$s = 0.1$	$s = 1$	$s = 10$	stable
disloyal Customer	0.050	0.070	0.076	✓
Inflight entertainment	0.128	0.171	0.181	✓
Class: Business	0.134	0.184	0.192	✓
Personal Travel	0.314	0.349	0.354	✓
Seat comfort	0.322	0.366	0.382	✓
On-board service	0.302	0.367	0.375	✓
Ease of Online booking	0.364	0.403	0.434	✓
Dep/Arr time convenient	0.363	0.419	0.431	✓
Checkin service	0.363	0.428	0.437	✓
Leg room service	0.394	0.444	0.462	✓
Food and drink	0.446	0.483	0.491	✓
Online support	0.467	0.515	0.525	
Cleanliness	0.498	0.541	0.555	
Class: Eco Plus	0.647	0.658	0.680	✓
Baggage handling	0.688	0.700	0.699	✓
log(1+Departure Delay)	0.686	0.702	0.715	✓
Inflight wifi service	0.704	0.709	0.716	✓
Online boarding	0.718	0.734	0.742	✓
Gate location	0.736	0.755	0.763	✓
Age	0.751	0.761	0.753	✓
Mid-flight delay (signed log)	0.762	0.762	0.791	✓
Flight Distance	0.754	0.763	0.768	✓

E. Feature selection cross-check with BAS

After fitting our feature-selection model based on a horseshoe prior, we would like to compare it with a Bayesian Adaptive Sampling (BAS) model in order to verify whether the two methods agree on which covariates are more useful in explaining the target variable.

With 23 predictors, $2^{23} \approx 8.4$ million candidate models are far too many to enumerate exhaustively, so we use `method = "MCMC"` in BAS to stochastically search the model space. A later section assesses whether this search suffers from instability or not.

Table 18 reports the marginal posterior inclusion probabilities $P(\beta_j \neq 0 | Y)$ for every predictor, together with the five highest-probability models returned by the search and their summary statistics (Bayes factor, posterior probability, R^2 , dimension and log marginal likelihood).

Table 18. Marginal posterior inclusion probabilities and the top five models identified by the BAS MCMC search. A value of 1.000/0.000 in a model column indicates the predictor is included/excluded in that model.

	$P(\beta \neq 0 Y)$	Model 1	Model 2	Model 3	Model 4	Model 5
Intercept	1.000	1.000	1.000	1.000	1.000	1.000
seat_comfort	0.803	1.000	1.000	1.000	1.000	0.000
time_convenient	0.973	1.000	1.000	1.000	1.000	1.000
food_drink	0.500	1.000	0.000	1.000	0.000	0.000
gate_location	0.037	0.000	0.000	0.000	0.000	0.000
wifi	0.049	0.000	0.000	0.000	0.000	0.000
entertainment	1.000	1.000	1.000	1.000	1.000	1.000
online_support	0.220	0.000	0.000	0.000	0.000	0.000
ease_booking	0.954	1.000	1.000	1.000	1.000	1.000
onboard_service	0.996	1.000	1.000	1.000	1.000	1.000
legroom	0.965	1.000	1.000	1.000	1.000	1.000
baggage	0.073	0.000	0.000	0.000	0.000	0.000
checkin	0.986	1.000	1.000	1.000	1.000	1.000
cleanliness	0.310	0.000	0.000	0.000	0.000	0.000
online_boarding	0.029	0.000	0.000	0.000	0.000	0.000
Age_std	0.037	0.000	0.000	0.000	0.000	0.000
FlightDist_std	0.035	0.000	0.000	0.000	0.000	0.000
Class_EcoPlus	0.029	0.000	0.000	0.000	0.000	0.000
Class_Business	0.993	1.000	1.000	1.000	1.000	1.000
logdep_std	0.092	0.000	0.000	0.000	0.000	0.000
TypeTravel_Personal	0.451	0.000	0.000	1.000	1.000	1.000
CustType_disloyal	1.000	1.000	1.000	1.000	1.000	1.000
logmid_delay	0.031	0.000	0.000	0.000	0.000	0.000
BF	NA	1.000	0.574	0.477	0.369	0.268
PostProbs	NA	0.128	0.071	0.059	0.048	0.033
R^2	NA	0.426	0.421	0.430	0.425	0.420
dim	NA	11.000	10.000	12.000	11.000	10.000
logmarg	NA	-516.838	-517.392	-517.578	-517.835	-518.155

F. Parameter-space convergence

We now test the goodness of the MCMC search over the parameter space executed by BAS.

Convergence within one run

BAS reports two independent estimates of each posterior inclusion probability:

- `probne0.RB`: takes the set of distinct visited models that contain variable j , uses their marginal likelihoods, renormalises and sums the mass of those models;
- `probne0.MCMC`: the raw fraction of iterations the chain spent in some model containing variable j .

If the chain has converged, the two estimates should agree closely.

The correlation between the two estimates is 0.9999 and the maximum absolute difference is 0.014: the points sit essentially on the diagonal (Figure 24), the sign that this run has converged.

Stability across seeds

We refit the identical model twice more with different random seeds and compare the posterior inclusion probabilities (PIPs) to check whether they agree. Table 19 lists the predictors with the largest spread, ordered by the range across the three seeds.

BIC prior as a robustness check

We try another prior on the β coefficients: if the search is stable, the effect of the prior should vanish once the number of iterations is large enough. Table 20 compares the g -prior PIPs with those obtained under a BIC-based prior.

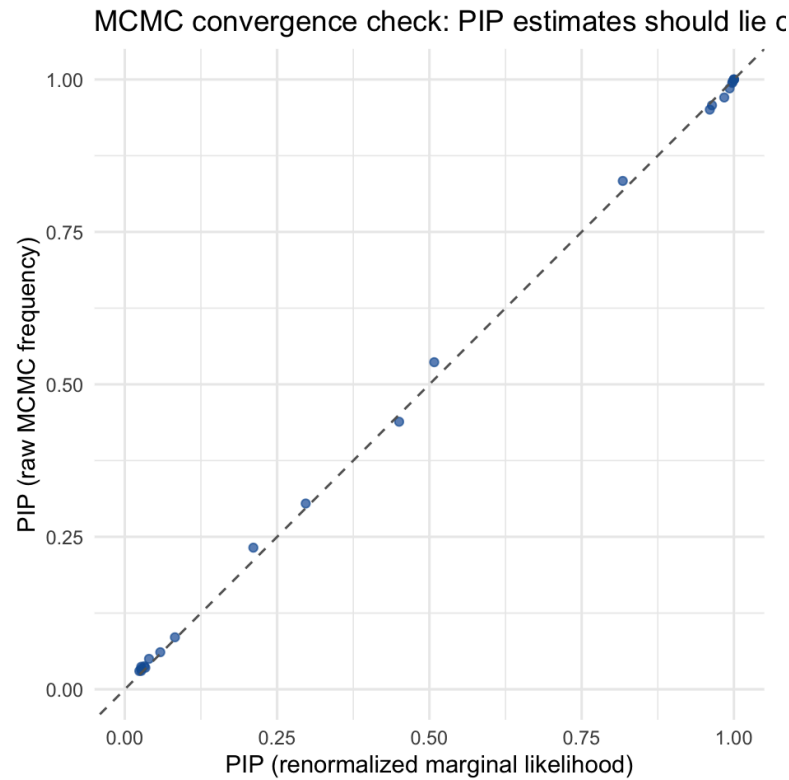


Figure 24. MCMC convergence check. Renormalised-marginal-likelihood inclusion probabilities (x-axis) against raw MCMC-frequency inclusion probabilities (y-axis); points lying on the dashed diagonal indicate the two estimates agree.

Table 19. Posterior inclusion probabilities across three random seeds, with the range (max – min) per predictor. Only the eight most variable predictors are shown.

term	seed123	seed1	seed2	range
seat_comfort	0.803	0.841	0.826	0.038
food_drink	0.500	0.533	0.511	0.033
online_support	0.220	0.228	0.204	0.024
cleanliness	0.310	0.297	0.316	0.019
TypeTravel_Personal	0.451	0.439	0.457	0.017
legroom	0.965	0.951	0.959	0.013
time_convenient	0.973	0.979	0.983	0.011
baggage	0.073	0.065	0.064	0.009

Table 20. Posterior inclusion probabilities under the g -prior and the BIC-based prior, with the absolute difference per predictor (six most affected predictors shown).

term	g_prior	BIC	abs_diff
online_support	0.220	0.224	0.004
cleanliness	0.310	0.314	0.004
seat_comfort	0.803	0.801	0.002
food_drink	0.500	0.498	0.002
ease_booking	0.954	0.952	0.002
legroom	0.965	0.967	0.002

With 100,000 MCMC iterations, PIPs move by at most 0.038 across the three seeds and by at most 0.004 between the g -prior and the BIC-based prior. Given the number of MCMC iterations, the search proves itself to be stable.

G. BAS vs. Horseshoe: model comparison

Finally, we compare the models selected by the two approaches and check on which coefficients they agree and on which they do not. Figure 25 plots each predictor's BAS inclusion probability against the horseshoe's $1 - \kappa_j$ shrinkage measure, colouring predictors by whether the two methods make the same signal/noise call.

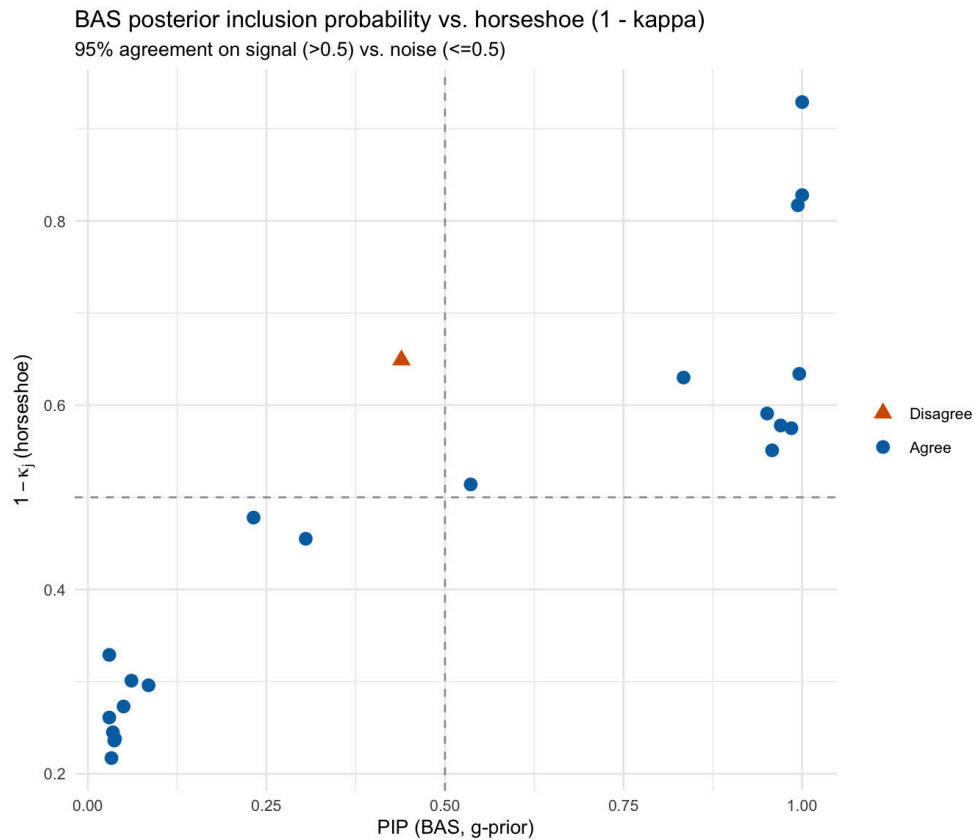


Figure 25. BAS posterior inclusion probability (x-axis) against the horseshoe's $1 - \kappa_j$ (y-axis). Dashed lines mark the 0.5 signal/noise cutoff for each method; green points denote agreement, the red point denotes the single disagreement.

91% of the 22 predictors get the same signal/noise call from BAS's model-space search and from the horseshoe's continuous shrinkage – two structurally different estimation strategies (discrete model averaging vs. a single shrinkage prior) landing on the same handful of variables. The two disagreements are genuine borderline cases in both methods: `Personal Travel` (PIP 0.439, $\kappa = 0.353$) sits just on opposite sides of each method's 0.5 cutoff, and `Food and drink` (PIP ≈ 0.50 , $\kappa = 0.489$) sits essentially on the cutoff for both methods and flips across MCMC seeds (Table 19) – not a real contradiction between the two approaches, but two variables where the signal is genuinely too weak to classify with confidence.

References

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